

***Science* Acronyms and Abbreviations**

Notes:

sft = define first time. (Sometimes it's less awkward to give the acronym first and then the spelled-out form in parentheses.) If a sft-type acronym is used only once or twice, use only the spelled-out form each time, unless the acronym is better known than the spelled-out form.

For entries with **alternate final word(s)** separated by a comma (e.g., “microscope, microscopy”), the acronym should be used to denote one spelled-out form or the other, but not both, in a given text.

Alternate forms of an acronym in the same entry indicate that more than one form is in current use.

<i>A</i>	absorbance (of light by solutions); see also OD	sft
<i>A</i> ₂₆₀	absorbance at 260 nm	sft
aa	amino acid	Spell out
aa-tRNA	aminoacyl-tRNA	sft
AA	arachidonic acid	sft
AARS	aminoacyl-tRNA synthetase	sft
A β	amyloid- β (Alzheimer's disease peptide)	sft
Ab	antibody	sft
ABC	ATP-binding cassette	sft
ac	alternating current	OK as is
ACC	anterior cingulate cortex	sft
ACh	acetylcholine	sft
AChE	acetylcholinesterase	sft
AChR	acetylcholine receptor	sft
AcMNPV	<i>Autographa californica</i> nuclear polyhedrosis virus	sft
AcOH	acetic acid (CH ₃ COOH)	sft
ACP	acyl carrier protein (domain)	sft
ACS	acetyl-coenzyme A synthase	sft
ACSF	artificial cerebrospinal fluid	sft
ACTH	adrenocorticotrophic hormone	sft (do not use ACTH ₄₋₁₀ or ¹⁻³⁸ ACTH)
Ad, Ad2	adenovirus; adenovirus type 2 or adenovirus-2	sft
ADA	adenosine deaminase	sft
AdoHCy	<i>S</i> -adenosylhomocysteine	Spell out
ADP	adenosine 5'-diphosphate	sft (5'- may be omitted)
ADPR	adenosine 5'-diphosphoribose	sft
AEV	avian erythroblastosis virus	sft
AFM	atomic force microscopy, microscope	sft
Ag	antigen	sft
AGB	asymptotic giant branch (star)	sft
AGCM	atmospheric general circulation model	sft
AH, AHP	after-hyperpolarization	Spell out
AIB	α -aminoisobutyric acid	sft
AIDS	acquired immunodeficiency syndrome	OK as is
AIF	apoptosis-inducing factor	sft
AK	adenosine kinase	sft
AKAP	adenosine kinase-associated protein	sft
Akt	(a serine-threonine protein kinase)	OK as is
ALA	δ -aminolevulinic acid	sft
ALLM	<i>N</i> -acetyl-Leu-Leu-methioninal	sft
ALLN	<i>N</i> -acetyl-Leu-Leu-norleucinal	sft
ALV	avian leukosis virus	sft
a.m.	ante meridiem	OK as is (usually use 24-hour time)

AMD	actinomycin D	sft
AMP	adenosine 5'-monophosphate	sft (5'- may be omitted)
AMPA	α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid	OK as is
AMP-PNP	adenylyl-imidodiphosphate	sft
AMOVA	analysis of molecular variance	sft
AMS	accelerator mass spectrometry	sft
ANF	atrial natriuretic factor	sft
ANCOVA	analysis of covariance	sft
ANOVA	analysis of variance	sft
ANP	atrial natriuretic peptide	sft
ANPP	aboveground net primary production	sft
AOM	acousto-optic modulator	sft
AP	alkaline phosphatase	sft
4AP	4-aminopyridine	sft
AP1	activating protein 1 (a transcription factor)	sft
APC	anaphase-promoting complex	sft
APC/C	anaphase-promoting complex/cyclosome	sft
APCs	antigen-presenting cells (usually plural)	sft
APMA	<i>p</i> -aminophenylmercuric acetate	sft
apoA, apoB, apoB-100, apoE	apolipoprotein A, apolipoprotein B, apolipoprotein B-100, apolipoprotein E	sft
APP	amyloid- β precursor protein	sft
APTES	3-aminopropyltriethoxysilane	sft
APUD	amine precursor uptake and decarboxylation	sft
APV	D,L-2-amino-5-phosphonovaleric acid	sft
AQDS	anthraquinone-2,6-disulfonate	sft
β_1 AR, β_2 AR	β_1 -adrenergic receptor, β_2 -adrenergic receptor	sft
Ara	arabinosyl, arabinose	sft
Ara-A	arabinosyladenine	sft
Ara-C	arabinosylecytosine (not cytosine arabinoside)	sft
Ara-H	arabinosylhypoxanthine	sft
ARE	AU-rich element	sft
ARF	ADP ribosylation factor (family of GTPases)	sft
ARM	anhysteretic remanent magnetization	sft
ARPES	angle-resolved photoemission spectroscopy; angle-resolved photoelectron spectroscopy	sft
ASK	apoptosis signal-regulating kinase	sft
ASO	allele-specific oligonucleotide	sft
ASV	avian sarcoma virus	sft
ASV	anodic stripping voltammetry	sft
ATCC	American Type Culture Collection	sft
ATF1, ATF2	activating transcription factor 1, 2	sft
ATM	ataxia-telangiectasia mutated	

ATP	adenosine 5'-triphosphate	sft (5'- may be omitted)
ATP- γ -S	adenosine 5'-O-(3-thiotriphosphate)	sft
ATPase	adenosine triphosphatase	sft (see Na ⁺ ,K ⁺ -ATPase)
AVHRR	advanced very-high-resolution radiometer	sft
AZT	azidothymidine; 3'-azido-3'-deoxythymidine, ; zidovudine (generic name): all three forms are OK	sft

$\beta\gamma$	(the $\beta\gamma$ heterodimer of G proteins)	Ensure that meaning is clear from context
B ₁₂	vitamin B ₁₂	Use “vitamin” first time
BAC	bacterial artificial chromosome	sft
BALB/c	(a strain of mouse)	OK as is
Balb/c	(a cell line)	OK as is
Balb/3T3	(a cell line)	OK as is
BaP, BP	benzo[<i>a</i>]pyrene	sft
BAPTA	1,2-bis(2-aminophenoxy)ethane- <i>N,N,N',N'</i> -tetraacetic acid (a Ca ²⁺ buffer)	sft
bcc	body-centered cubic	sft
BCE	Before the Common era [with a date] 399 BCE	OK as is
BCG	Bacille Calmette-Guérin (vaccine type)	sft
BCIP-NBT	bromochloroindolyl phosphate–nitro blue tetrazolium	sft
BCR	B cell antigen receptor, B cell receptor, B lymphocyte antigen receptor	sft (follow author)
BCS	Bardeen-Cooper-Schrieffer (superconductivity)	sft
B-DNA	B-form DNA (also non-B DNA)	sft
BDNF	brain-derived neurotrophic factor	sft
BdU	(use BrdU)	
BEC	Bose-Einstein condensate, condensation	sft
BEEM	ballistic electron emission microscopy	sft
BEMM	ballistic electron magnetic microscopy	sft
BESM	bovine embryo skeletal muscle	sft
BET	Brunauer-Emmett-Teller (adsorption isotherm)	sft
B/F	ratio of bound to free ligand	sft
BFA	brefeldin A	sft
bFGF	basic fibroblast growth factor	sft
BFU	burst-forming unit	sft
BGG	bovine gamma globulin	sft
α -bgt	α -bungarotoxin	sft
BHK cells	baby hamster kidney cells	sft
bHLH	basic helix-loop-helix	sft
BMDMs	bone marrow–derived macrophages (usually plural)	sft
BMMCs	bone marrow–derived mast cells (usually plural)	sft
BMP	bone morphogenetic protein	sft
BMR	basal metabolic rate	sft
BOC, Boc	butoxycarbonyl	sft
BOD	biological oxygen demand	sft
BOLD	blood oxygen level–dependent	sft
B.P.	before the present (used after unit of time)	sft
bp	base pair (as unit after a number)	sft (but spell out when not used as

		unit, e.g., “base pairs 50 to 85”)
BPTI	bovine pancreatic trypsin inhibitor	sft
BR	bacteriorhodopsin	sft
BrdU	5-bromo-2'-deoxyuridine	sft (5- and 2'- may be omitted)
Br-UTP	bromouridine 5'-triphosphate	sft
BSA	bovine serum albumin	sft
BSE	backscattered electron (imaging)	sft
BSE	bovine spongiform encephalopathy	sft
BSF	B cell–stimulating factor	sft
BSO	buthionine sulfoximine	sft
BSS	balanced salt solution	sft
B-tropic	(virus that grows better on Balb/c cells)	sft
Btk	Bruton’s tyrosine kinase	sft
Bu	butyl	sft
ⁿ Bu	<i>n</i> -butyl	sft
BUN	blood urea nitrogen	sft
Bz	benzyl	sft

C	complement	sft
C	constant regions of immunoglobulin chains	OK as is
C terminal, C terminus	carboxyl-terminal, carboxyl terminus	Spell out in abstract; otherwise OK as is
C ₃ plant	first product of photosynthesis is a 3-carbon acid (e.g., phosphoglycerate)	Ensure that meaning is clear from context, e.g., “plants using the C ₃ photosynthetic pathway”
C ₄ plant	first product of photosynthesis is a 4-carbon acid (e.g., oxaloacetate)	Ensure that meaning is clear from context, e.g., “plants using the C ₄ photosynthetic pathway”
C ₁₈ chain	chain containing 18 carbon atoms	OK as is
C _H	constant portion of the immunoglobulin heavy chain	sft
C _{H1}	domain 1 of C _H	sft
C α	constant portion of the α chain of a T cell receptor	Ensure that meaning is clear from context
C1, C2 . . . C9	components of complement	sft
C3, C4	types of photosynthesis	OK as is
C5a	fragment of complementary protein C5 (a complement-derived anaphylatoxin; glycoprotein fragment cleaved from C5)	sft
C57BL	a strain of mouse	OK as is
C-18	carbon at position 18	OK as is
(C18:2)	for unsaturated (with double bond) fatty acids, ratio of C atoms to double bonds	sft
1C, 2C	a gamete has 1C amount of DNA, a diploid G ₁ cell has 2C amount of DNA	sft
¹⁴ C yr B.P.	radiocarbon years before the present	sft
[Ca ²⁺] _i	intracellular calcium concentration	sft
CA1, CA3, CA4	(hippocampal regions)	OK as is
CAD MS	collisionally activated dissociation mass spectrometry	sft
CAEV	caprine arthritis encephalitis virus	sft
CAIs	calcium-aluminum-rich inclusions	sft
CALI	chromophore-assisted laser activation	sft
CAM	crassulacean acid metabolism	sft
CAM	chorioallantoic membrane	sft
CAM	cell adhesion molecule	sft

CaM	calmodulin (a protein that binds to Ca ²⁺)	sft
CaMKII, CaMKIV	calcium- and calmodulin-dependent protein kinase II, IV	sft
cAMP	adenosine 3',5'-monophosphate (if AMP has not yet been defined); cyclic AMP (if AMP has already been defined) [Author may prefer "cyclic adenosine monophosphate," also OK]	sft
[cAMP];	intracellular cAMP concentration	sft
CaMV	cauliflower mosaic virus	sft
CAR	constitutive androstane receptor (a nuclear receptor)	sft
CAR T cell	chimeric antigen receptor T cell	sft
CARS	continuous anti-Stokes Raman scattering (or spectroscopy)	sft
CAT	chloramphenicol acetyltransferase	sft
CAT	computerized axial tomography	sft
CBF	cerebral blood flow	sft
CBP	CREB-binding protein	sft
C cell	calcitonin-producing parafollicular cell (in thyroid)	sft
CCN	cloud condensation nuclei	sft
CCR1, CCR5, CCR6	(human chemokine receptors)	OK as is
cDNA	complementary DNA	Define only in abstracts
ccDNA	closed circular DNA	sft
cccDNA	covalently closed circular DNA	sft
CCD	charge-coupled device	sft
CCK	cholecystokinin	sft
CD	circular dichroism	sft
CD4, CD8, CD34	(antigens)	OK as is
Cdc24	(a yeast GEF for Cdc42)	OK as is
Cdc42	(a small GTPase)	OK as is
CDK	cyclin-dependent kinase; specific ones Cdk1, Cdk2, etc.	Sft
CDP	cytidine 5'-diphosphate	sft (5'- may be omitted)
CDTA	cyclohexane diamine tetraacetic acid	Sft
CDW	charge density wave	Sft
CE	Common era [with a date] 2013 CE	OK as is
CEA	carcinoembryonic antigen	sft
C/EBP	(a transcription factor, CCAAT/enhancer binding protein)	sft
CEC	chick embryo cells	sft
CEF	chick embryo fibroblast	sft
CEPH	Centre d'Etude du Polymorphisme Humain	sft

CER	CO ₂ exchange rate	sft
Cer	ceramide	Spell out
Cer Gal	galactocerebroside (cerebral galactoside)	Check with author
Cer Glc	glucocerebroside	Check with author
CF	cystic fibrosis	sft
CFA	complete Freund's adjuvant	Change to FCA, Freund's complete adjuvant
<i>c-fos</i>	(a transcription factor proto-oncogene)	OK as is
<i>c-jun</i>	(a transcription factor proto-oncogene)	OK as is
c-Myc	(a transcription factor)	OK as is
CFP	cyan fluorescent protein	sft
CFSE	carboxyfluorescein diacetate succinimidyl ester	sft
CFTR	cystic fibrosis transmembrane conductance regulator	sft
CFU	colony-forming units (“s” generally not required for plural, but if both individual and plural are mentioned, use CFU/CFUs)	sft
CFU-E	erythroid colony-forming units	sft
CG	chorionic gonadotropin (see hCG)	sft
cGMP	guanosine 3',5'-monophosphate (if GMP has not yet been defined); cyclic GMP (if GMP has already been defined) [Author may prefer “cyclic guanosine monophosphate,” also OK]	sft
CGSEL	circular grating–coupled surface-emitting laser	sft
CHD	1,3-cyclohexadiene	sft
ChAT	choline acetyltransferase	sft
CHAPS	3-[(3-cholamidopropyl)dimethylammonio]-1- propane sulfonate	OK as is
ChE	cholinesterase	sft
ChIP	chromatin immunoprecipitation	sft
chlorophyll a,b,c,d,a/b	(roman letters)	OK as is
CHO	Chinese hamster ovary (cells)	sft
CHX	cycloheximide	sft
CI	confidence interval (usually preceded by 90% or 95%)	sft
CI	cytotoxicity index	sft
CID	collision-induced dissociation	sft
CIP	calf intestinal phosphatase	sft
<i>cis</i> (and <i>trans</i>)	(isomers, italic before a specific compound, roman before a group term)	OK as is
<i>cis</i> (in genetics)	acts in a <i>cis</i> manner (for example, <i>Gur</i> affects only the <i>Gus</i> gene on the same chromosome); <i>cis</i> activity; in <i>cis</i>	OK as is (leave in roman type)
CJD	Creutzfeldt-Jakob disease	sft

CKI	casein kinase I	sft
CKI	CDK inhibitor	Spell out to avoid confusion with casein kinase I
CLIP	class II–associated invariant chain peptides	sft
Clip	corticotropin-like intermediate lobe peptide	sft
c.m.	center of mass (lowercase)	sft
CM	carboxymethyl	sft
CMB	core-mantle boundary	sft
CMBT	5-chloro-2-mercaptobenzothiazole	sft
CMC	<i>O</i> -carboxymethylcellulose	sft
CMOS	complementary metal-oxide semiconductor	sft
CMP	cytidine 5′-monophosphate; sometimes called cytidylic acid	sft (5′- may be omitted)
CMR	colossal magnetoresistance	sft
CMR	cerebral metabolic rate	sft
CMRO ₂	cerebral metabolic rate for O ₂	Spell out
CMV	cytomegalovirus	sft
CN	cyanide	OK as is
CNQX	6-cyano-7-nitroquinoxaline-2,3-dione	sft
CNS	central nervous system	sft
CNTF	ciliary neurotrophic factor	sft
CNX	calnexin	sft
CoA	coenzyme A	sft
CODH	carbon monoxide dehydrogenase	sft
COMT	catechol- <i>O</i> -methyltransferase	sft
Con A	concanavalin A	sft
COPI, COPII	coat protein complex I, II	sft
COS, COS-7	(a cell line)	OK as is
COSPEC	correlation spectrometry	sft
COSY	correlation spectroscopy	sft
COVID-19	coronavirus disease 2019	OK as is
COX	cyclooxygenase	sft
CPB	competitive protein binding	sft
CPD	citrate phosphate dextrose	sft
CPE	cytopathic effect	sft
CP/MAS NMR	cross-polarization/magic angle spinning NMR	sft
CPP	carboxypiperazin-4-yl-propyl-1-phosphonic acid	sft
8-CPT cAMP	8-(4-chlorophenylthio)-cAMP	sft (spell out cAMP if necessary)
CPY	carboxypeptidase Y	sft
CRD	cross-reacting determinant	sft
CRE	cyclic AMP (or cAMP) response element	sft
CRE	cosmic ray exposure (modifies age or data)	sft
CREB	cAMP response element–binding protein	sft
CREM	cAMP response element modulator	sft

CRF	corticotropin-releasing factor	sft
CRH	corticotropin-releasing hormone	sft
CRISPR	clustered regularly interspaced short palindromic repeats	OK as is
CRM	cross-reacting material	sft
CRM	chemical remanent magnetization	sft
cRNA	complementary RNA	sft
CRP	complement-reactive protein	sft
CRT	cathode ray tube	sft
CRT	calreticulin	sft
cryo-EM	cryo-electron microscopy	sft
cryo-TEM	cryo-transmission electron microscopy (preferred over CTEM, cryotransmission electron microscopy)	sft
CsA	cyclosporin A	sft
CSBP-1	(a p38 mitogen-activated protein kinase, also known as p38; this can be confusing)	Ensure that meaning is clear from context
CSF	cerebrospinal fluid	sft
CSF	colony-stimulating factor	sft
CSF	cytostatic factor	sft
CSF	classical swine fever (virus)	sft
CSGE	conformation-sensitive gel electrophoresis	sft
Csk	C-terminal Src kinase	sft
CSP	circumsporozoite precipitation	sft
CT0, CT12	subjective dawn, subjective dusk (circadian convention)	sft
CTD	conductivity-temperature-depth (profile)	sft
CTD	C-terminal repeat domain	sft
CTL	cytotoxic T lymphocyte	sft
CTP	cytidine 5'-triphosphate	sft (5'- may be omitted)
CU	calliphora unit = 0.01 g of ecdysone (unit of ecdysone in insects)	sft
CV	conventional flora	sft
cv	cultivar	Spell out (unless used repeatedly)
CV	curriculum vitae [plural, curricula vitae (CVs)]	sft
CVD	chemical vapor deposition	sft
CW, cw	continuous wave (laser)	sft
CXCR4	(a human chemokine receptor)	OK as is
cytochromes a, b, and c	(do not italicize)	OK as is

Δ	geocentric distance, distance from Earth	sft
D	deuterium	OK as is
D''	(term for a discontinuity in Earth's mantle)	Ensure that meaning is clear from context
1D, 2D, 3D	one-, two-, three-dimensional (once one is defined, others can be used unexplained); but "3D printing" is acceptable without definition	sft
D-	indicates stereoisomerism (small cap) for amino acids & carbohydrates	Use small cap
d-	dextrorotatory	OK as is (but see IUB/IUPAC: they prefer +)
d	deoxy (carbohydrates and nucleotides) as in dATP, dCTP, dCMP, etc.	Spell out (see dATP, dCTP, etc.)
DAG	diacylglycerol	sft
D-ALA	[D-Ala ²]methionine enkephalinamide	sft (small cap D)
DAPI	4',6-diamidino-2-phenylindole	sft [prime symbol over 4 only]
D-APV	D-aminophosphovalerate	sft
dATP	2'-deoxyadenosine 5'-triphosphate	OK as is if ATP has previously been defined; otherwise, sft (2'- and 5'- may be omitted)
DBH	diameter at breast height (forestry)	sft
DBS	dodecylbenzene sulfonate	sft
dc	direct current	OK as is
DCC	dicyclohexylcarbodiimide	sft
D cells	delta cells of pancreas	sft
DCF	2,7-dichlorofluorescein diacetate	sft
DCM	dichloromethane	sft
dCMP	2'-deoxycytidine 5'-monophosphate	OK as is if CMP has previously been defined; otherwise, sft (2'- and 5'- may be omitted)
dCTP	2'-deoxycytidine 5'-triphosphate	OK as is if CTP has previously been defined; otherwise, sft (2'- and 5'- may be omitted)
dd	dideoxy (carbohydrates and nucleotides) as in ddATP, ddCTP, ddCMP, etc.—spelled-out forms same as the deoxy forms, but with "di" prefix	sft
DDC	2',3'-dideoxycytidine	sft

DDT	1,1,1-trichloro-2,2-bis(<i>p</i> -chlorophenyl)ethane	OK as is
DEAD,	Asp-Glu-Ala-Asp amino acid motif (shared by	OK as is
DEAD-box	helicase proteins that unwind nucleic acids)	
DEAE-	<i>O</i> -(diethylaminoethyl) cellulose	OK as is
cellulose		
3-deaza-Ado	3-deazaadenosine	sft
2DEG	two-dimensional electron gas	sft
DEGS	diethylene glycol succinate	sft
DEP	dielectrophoresis	sft
DEPC	diethyl pyrocarbonate	sft
DES	diethylstilbestrol	sft
DETCs	dendritic epidermal T cells (usually plural)	sft
DEXA	dual-energy x-ray absorptiometry	sft
df	degrees of freedom (statistics)	OK as is
DFI	differential fluorescence induction	sft
DFMO	difluoromethylornithine	sft
DFP	diisopropylphosphorofluoridate	OK as is
DFT	density functional theory	sft
2-DG	2-deoxyglucose	sft
DGGE	denaturant gradient gel electrophoresis	sft
dGTP	2'-deoxyguanosine 5'-triphosphate	OK as is if GTP has previously been defined; otherwise, sft (2'- and 5'- may be omitted)
DHCC	3,3'-dihexyloxacarbocyanine	sft
DHDPS	dihydrodipicolinic acid synthase	sft
DHFR	dihydrofolate reductase	sft
DHFR-TS	dihydrofolate reductase–thymidylate synthase	sft
DHPLC	denaturing high-performance liquid chromatography	sft
DHT	dihydrotestosterone	sft
DIC	dissolved inorganic carbon	sft
DIC	differential interference contrast	sft
dI-dC	deoxyinosine-deoxycytidine	sft
DiI	1,1'-dioctadecyl-3,3,3',3'- tetramethylindocarbocyanine perchlorate	sft
DIP	dissolved inorganic phosphate	sft
DISC	death-inducing signaling complex	sft
DIV	day(s) in vitro	sft
DL-	racemic (a mixture of dextro- and levo-)	OK as is
DLPFC	dorsolateral prefrontal cortex	sft
DLS	dynamic light scattering	sft
DMA	dimethylamine	sft
DMAA	dimethylallylamine	sft
DMBA	dimethylbenz[<i>a</i>]anthracene	sft

DMEM	Dulbecco's minimum essential medium, or Dulbecco's modified Eagle's medium	sft; query author
DMF	<i>N,N'</i> -dimethylformamide	sft
DMOG	dimethyloxalylglycine	sft
DMPA	dimyristoylphosphatidic acid	sft
DMPEA	dimethoxyphenylethylamine	sft
DMPH ₄	6,7-dimethyl-5,6,7,8-tetrahydropterine	sft
DMS	dimethyl sulfate or dimethyl sulfide	sft
DMSO	dimethyl sulfoxide	sft
DMT	4,4'-dimethoxytrityl	sft
DNA	(no need to spell out)	OK as is
DNA-PK	DNA-dependent protein kinase	sft
DNase	deoxyribonuclease	sft
DNP-	2,4-dinitrophenyl-	sft
dNTP	deoxynucleotide triphosphate, deoxynucleoside triphosphate	sft (query if undefined)
DOAS	differential optical absorption spectroscopy	sft
DOC	deoxycholate; 11-deoxycorticosterone	sft
DOC	dissolved organic carbon	sft
DO depletion	dissolved oxygen depletion	sft
DON	6-diazo-5-oxo-L-norleucine	sft (small cap L)
dopa	3,4-dihydroxyphenylalanine	sft
DOPAC	3,4-dihydroxyphenylacetic acid	sft
dopamine	3,4-dihydroxyphenylethylamine	OK as is
DOPC	dioleoylphosphatidylcholine	sft
DOPG	dioleoylphosphatidylglycerol	sft
DOS	density of states	sft
DPC	diphenylamine carboxylate	sft
DPG	diphosphoglycerate	sft
DPV	differential pulse voltammetry	sft
DQC	double quantum coherence	sft
DQF-COSY	double quantum-filtered correlation spectroscopy	sft
DR1	disperse red-1 (a chromophore)	sft
DRE	dense rock equivalent	sft
DRGs	dorsal root ganglia (usually plural)	sft
DRLM	depolarized reflected light microscopy	sft
dRP	deoxyribose phosphate	sft
DSB	double-strand break	sft
DSC	differential scanning calorimetry	sft
dsDNA	double-stranded DNA	sft
dsRNA	double-stranded RNA	sft
DSS	dextran sulfate sodium	sft
DTH	delayed-type hypersensitivity	sft
dTTP	3'-deoxythymidine 5'-triphosphate	OK as is if TTP has previously been defined; otherwise,

DTPA	diethylenetriamine pentaacetic acid	sft (3'- and 5'- may be omitted)
DTR	diurnal temperature range	sft
DTT	dithiothreitol	sft

e^-	electron (use when referring to electron particles)	sft
e	charge on the electron	sft
e	natural base (approximately 2.7183)	OK as is
e^x	exponential of x [$\exp(x) = e^x$]; see exp	OK as is
E	embryonic day [usually introduce with number: “E11 (embryonic day 11)”]	sft
E_0	(sub zero) standard redox potential, pH 0, 25°C	sft
E_0'	(sub zero, stack prime) same as E_0 at pH 7, standard redox potential [corrected]	sft
E_2	17 β -estradiol or estradiol-17 β	sft (use 17 β -estradiol)
E_{Na}	electrochemical equilibrium potential for sodium	sft
E_R	resting potential	sft
E_r, E_{red}	see E_0 above	Avoid
E1A	early region 1A proteins of adenovirus	sft
EAA	excitatory amino acid	sft
EBV	Epstein-Barr virus	sft
EC ₅₀	median effective concentration	sft
ECG	electrocardiography; electrocardiogram	sft
ECGC	electron-capture gas chromatography	sft
ECL	enhanced chemiluminescence	sft except when used with “(Amersham)”
ECM	extracellular matrix	sft
E.C. number	Enzyme Commission number	OK as is
ECW	extracellular water	sft
EDDHA	ethylenediamine di-(<i>O</i> -hydroxyphenylacetate)	sft
ED ₅₀	median effective dose or half-maximal effective dose	sft
EDC, EDAC	1-ethyl-3-(3-dimethylaminopropyl)carbodiimide	sft
EDHF	endothelium-derived hyperpolarizing factor	sft
EDM	electronic distance measurement	sft
EDTA	ethylenediaminetetraacetic acid	OK as is
EDX	energy-dispersive x-ray (spectroscopy, or analysis)	sft
EDXD	energy-dispersive x-ray diffraction	sft
EEG	electroencephalography; electroencephalogram	sft
EELS	electron energy-loss spectroscopy	sft
EEPROM	electrically erasable programmable read-only memory	sft
EFM	electrostatic force microscopy	sft
EF-Tu	elongation factor Tu	sft
EG	ethylene glycol	sft
EGF	epidermal growth factor	sft
EGFR	epidermal growth factor receptor	sft
EGFP, eGFP	enhanced green fluorescent protein	sft
EGME	ethylene glycol monoethyl ether	sft

EGTA	ethyleneglycol-bis-(β -aminoethyl ether)- N,N,N',N' -tetraacetic acid	OK as is
EHD	electrohydrodynamic	sft
EIA	enzyme immunoassay	sft
EI-MS	electron impact–mass spectrometry	sft
EKG	electrocardiography; electrocardiogram	(use ECG)
ELDOR	electron-electron double resonance	sft
ELISA	enzyme-linked immunosorbent assay	sft
EM	electron microscopy; electron microscope	sft
EM cells	endostial marrow cells	sft
EMCV	encephalomyocarditis virus	sft
EMF	electromotive force	sft
EMG	electromyogram, electromyography	sft
EMS	ethyl methanesulfonate	sft
EMSA	electrophoretic mobility shift assay	sft
EN	ethylenediamine	sft
ENDOR	electron nuclear double-resonance (spectroscopy)	sft
ENS	elastic neutron scattering	sft
ENSO	El Niño–Southern Oscillation (initial caps)	sft
ENU	N -ethyl- N -nitrosourea	sft
β -EP	β -endorphin	sft
EPMA	electron probe microanalysis	sft
EPR	electron paramagnetic resonance	sft (same as ESR)
EPSC	excitatory postsynaptic current	sft
EPSP	excitatory postsynaptic potential	sft
Eq., Eqs.	equation, equations	OK to use with number
ER	endoplasmic reticulum	sft
ERBE	Earth Radiation Budget Experiment	sft
ERK	extracellular signal–regulated kinase (a type of MAPK, per)	sft
ERP	event-related potential	sft
ES	embryonic stem (cell line)	sft
ESEEM	electron spin echo envelope modulation	sft
ESI	electrospray ionization	sft
ESI-MS	electrospray ionization mass spectrometry	sft
ES-MS	electrospray mass spectrometry	sft
ESR	electron spin resonance	sft (same as EPR)
EST	expressed sequence tag	sft
Et	ethyl	sft
EtOH	ethanol	sft
EXAFS	extended x-ray absorption fine structure	sft
execution point	when 50% of cells enter the S phase of the cell cycle	sft
exp(x)	exponential of x [$\exp(x) = e^x$]; see e^x	use e^x

F_1, F_2, F_3	filial generation (genetics)	OK as is
$f/64$	(photographic aperture ratio)	OK as is
$F_{\text{obs}}, F_{\text{calc}}$	(observed and calculated structure factors) (use instead of F_o, F_c)	sft; query; varies by manuscript
FAAH	fatty acid amide hydrolase	sft
FAB	fast atom bombardment	sft
Fab	antigen-binding fragment	OK as is
FAB-MS	fast atom bombardment mass spectrometry	sft
FABP	fatty acid-binding protein	sft
FACS	fluorescence-activated cell sorting, sorter	sft
FAD	flavin adenine dinucleotide	sft
FADD	Fas-associated death domain protein	sft
FAE	follicle-associated epithelium	sft
FAH	fumaryl acetoacetate hydrolase	sft
FAK	focal adhesion kinase	sft
FALDH	fatty aldehyde dehydrogenase	sft
FAS	fatty acid synthase	sft
FBS	fetal bovine serum (preferred to FCS, fetal calf serum)	sft
Fc	crystalline fragment; an antibody (Ig) fragment from which C_H is crystallizable	OK as is
FCA	Freund's complete adjuvant	sft
fcc	face-centered cubic	sft
FCCP	carbonyl cyanide <i>p</i> -trifluoromethoxyphenylhydrazone	sft
FcR	Fc receptor	sft
FCS	fluorescence correlation spectroscopy	sft
FDA	fluorescein diacetate	sft
FDA	Food and Drug Administration	sft
FD&C	Food, Drug, and Cosmetic (Act)	OK as is
FDCs	follicular dendritic cells (usually plural)	sft
FDNB	1-fluoro-2,4-dinitrobenzene	sft
FDTD	finite-difference time-domain (calculations)	sft
fEBP	fetoneonatal estrogen-binding protein	sft
FEL	free-electron laser	sft
FELV, FeLV	feline leukemia virus	sft
FE-SEM, FESEM	field-emission scanning electron microscopy, microscope	sft
FET	field-effect transistor	sft
FF	fluorescent foci	sft
FFE	free-flow electrophoresis	sft
FFT	fast Fourier transform	sft
FFU	focus-forming units	sft
FGF	fibroblast growth factor	sft
FID	flame ionization detector	sft
FID	free induction decay (used in NMR)	sft

FIGE	field-inversion gel electrophoresis	sft
FIM	field ion microscopy	sft
FISH	fluorescence in situ hybridization	sft
FITC	fluorescein isothiocyanate	sft
FLAG, Flag	epitope DYKDDDDKV (usually seen as “FLAG epitope–tagged” or “FLAG-tagged”)	OK as is
FLICE	FADD-like interleukin-1 β –converting enzyme	sft
FLIM	fluorescence lifetime imaging microscopy	sft
FMet	formylmethionyl; formylmethionine	sft
fMLP	<i>N</i> -formyl-Met-Leu-Phe	sft (<i>italic f</i>)
FMN	riboflavin 5'-phosphate	sft
FMN	flavin mononucleotide	sft
FMNH	reduced form of FMN	sft
Fmoc	9-fluorenyl methoxycarbonyl	sft
fMRI	functional magnetic resonance imaging	sft
f_{O_2}	oxygen fugacity (small cap O, <i>italic f</i> ; Earth mantle papers)	sft
FOM	figure of merit	sft
FOV	field of view	sft
FPES	femtosecond photoelectron spectroscopy	sft
FPLC	fast protein liquid chromatography	sft
FRET	fluorescence resonance energy transfer, Förster resonance energy transfer [author preference]	sft
FRF	follicle-stimulating hormone–releasing factor (en dash)	sft
FRRF	fast repetition rate fluorometry	sft
FSH	follicle-stimulating hormone	sft
FSHRF	follicle-stimulating hormone–releasing factor (en dash)	sft (use FRF)
FTase	farnesyltransferase	sft
FTIR	Fourier transform infrared (spectroscopy, spectrometer)	sft
FTOC	fetal thymic organ culture	sft
5-FU	5-fluorouracil	sft
FW	formula weight	sft
FWHM	full width at half maximum	sft
Fv	heterodimer of the variable region of the light and heavy chains	sft
FXR	farnesoid X receptor (a nuclear receptor)	sft

ΔG°	Gibbs free energy (degree sign not superscripted)	sft
G ₀ , G ₁ , G ₂	(cell cycle notation: G ₀ , quiescent phase; G ₁ , prereplicative phase; G ₂ , period between S phase and onset of M phase)	Ensure that meaning is clear from context
G _{M1} , G _{D1} , G _T	(ganglioside with monosialic acid, disialic acid, or trisialic acid; 1, 2, 3 refer to terminal galactose missing)	OK as is
G6P	glucose-6-phosphate	sft
G6PD	glucose-6-phosphate dehydrogenase	sft
G418	(an antibiotic)	OK as is
G protein	heterotrimeric GTP-binding protein, heterotrimeric guanine nucleotide-binding protein [Author preference]	sft (but do not spell out GTP)
G ₁₂	(a family of G protein α subunits that includes G α_{12} and G α_{13})	
G α	(the α subunit of G proteins)	
G α_{12}	(the G protein α subunit that controls Rho GTPase activity)	
G α_{13}	(a G protein α subunit of the G ₁₂ family)	
G α_{i1} , G α_{i2} , G α_{i3}	(G protein α subunits of the G α_i family; see G _i)	
G α_o	(a G protein α subunit of the G α_i family; see G _i)	
G $\beta\gamma$	(the $\beta\gamma$ heterodimer of G proteins)	Ensure that meaning is clear from context
G _i	(the family of G protein α subunits that can inhibit adenylyl cyclases)	Ensure that meaning is clear from context
G _q	(the family of G protein α subunits that controls the activity of phosphatidylinositol-specific phospholipases)	Ensure that meaning is clear from context
G _s	(the family of G protein α subunits that can stimulate adenylyl cyclases)	Ensure that meaning is clear from context
Ga	10 ⁹ years (ago), billion years (ago)	sft
	[Commonly used to represent age. “Ago” is often implied but may be written out (“xx billion years ago”) when context is needed (follow authors).]	
GABA	γ -aminobutyric acid	sft
GABA _A	γ -aminobutyric acid type A	sft
GABAergic	GABA-releasing, GABA-mediated, GABA-potentiating, GABA-mimicking, etc.	Ensure that meaning is clear from context
GABA-T	GABA aminotransferase	Spell out
GAD	glutamic acid decarboxylase	sft
Gal	galactosyl, galactose	sft

β -Gal	β -galactosidase	sft
GalN	galactosamine	sft
GalNAc	<i>N</i> -acetylgalactosamine	sft
GALT	gut-associated lymphoid tissue	sft
GalTase	β -1,4-galactosyltransferase	sft
GALV	gibbon ape leukemia virus	sft
GAP	GTPase activating protein	sft (spell out GTPase first time)
GAPDH	glyceraldehyde-3-phosphate dehydrogenase	sft (-3- may be omitted)
GC	germinal center	sft
GC	gas chromatography, chromatographic	sft
GCI	glial cytoplasmic inclusions	sft
GCM	general circulation model	sft
GCM	glia-conditioned medium	sft
GC-MS	gas chromatography–mass spectrometry (en dash)	sft
GCR	galactic cosmic ray	sft
GCRP	G-coupled cell surface receptor protein	sft
G-CSF	granulocyte colony-stimulating factor	sft
GDF	growth and differentiation factor	sft
GDI	guanine nucleotide dissociation inhibitor	sft
gDNA	genomic DNA	sft
GDNF	glial cell line–derived neurotrophic factor	sft
GDP	guanosine diphosphate	sft
GDP- β -S	guanosine 5'- <i>O</i> -(2'-thiodiphosphate)	sft
GEF	guanine nucleotide exchange factor	sft
GEMS	glass with embedded metal and sulfides (astrophysics) (treat as plural)	sft
GFAP	glial fibrillary acidic protein	sft
GFC	gel filtration chromatography	sft
GFP	green fluorescent protein	sft
GGA	generalized gradient approximation	sft
GH	growth hormone (somatotropin)	sft
GHF, GHRF, GRF	growth hormone–releasing factor (en dash)	sft
GHIH, GHIF	GH-release inhibiting hormone or factor (somatostatin)	sft
GID	grazing incidence diffraction	sft
GIF	growth hormone releasing–inhibiting factor (en dash)	(use GHIH)
GILT	gamma interferon–inducible lysosomal thiol reductase	sft
GIP	gastric inhibitory peptide	sft
GISP	Greenland Ice Sheet Project	sft
GISS	Goddard Institute for Space Studies (climate model)	sft
GIXD	grazing incidence x-ray diffraction	sft

GKAP	guanylate kinase associated protein	sft
GLAD	glancing angle deposition	sft
Glc	glucose	sft
GLC	gas-liquid chromatography	sft
GlcN	glucosamine	sft
GlcNAc	<i>N</i> -acetylglucosamine	sft
GM-CSF	granulocyte-macrophage colony-stimulating factor	sft
GMHT	genetically modified herbicide-tolerant	sft
GMP	guanosine 5'-monophosphate	sft (5'- may be omitted)
GMR	giant magnetoresistive effect	sft
GMST	global mean surface temperature	sft
GMT	Greenwich mean time	OK as is
GGBP	guanine nucleotide-binding protein	sft
GNERF	guanine nucleotide release factor	sft
GnRH	gonadotropin-releasing hormone	sft
α -GP	α -glycerophosphate	sft
GPAT	glycerol 3-phosphate acyltransferase	sft
GPC	gel permeation chromatography	sft
GPCR	G protein-coupled receptor	sft
GPDH	glycerophosphate dehydrogenase	sft
GPI	glycosylphosphatidylinositol	sft
GPS	Global Positioning System (initial caps)	Ensure meaning is clear from context
GRF	growth hormone-releasing factor (en dash)	sft
GRK	G protein-coupled receptor kinase	sft
GRIP	Greenland Ice Core Project	sft
gRNA	guide RNA	sft
GroEL, GroES	(chaperonins)	OK as is
GS	group specific (hyphenate before a noun)	sft
GSCs	germline stem cells (usually plural)	sft
GSH	glutathione (reduced form)	sft
GSK	glycogen synthase kinase	sft
GSSG	oxidized glutathione	sft
GST	glutathione <i>S</i> -transferase	sft
GTP	guanosine 5'-triphosphate	sft (5'- may be omitted)
GTPase	guanosine triphosphatase	sft
GTP- γ -S	guanosine 5'- <i>O</i> -(3'-thiotriphosphate)	sft
GTT	glucose tolerance test	sft
GuHCl	guanidine hydrochloride	sft
GVHD	graft-versus-host disease	sft
GWAS	genome-wide association study (plural form is GWASs)	sft

Gyr	10^9 years, billion years [Commonly used to represent duration/length of time. Phrases may also be written out (“xx billion years”).]	sft; may use multiples of 1000 Myr instead
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$\hbar$	Planck's constant h divided by 2π	sft
H_0	Hubble constant	sft
H2B	histone 2B	sft
ΔH	change in enthalpy (thermodynamics variable)	sft
HI, HII, etc.	ionization states of hydrogen (astrophysics)	sft
HA	hemagglutinin (often as epitope: "HA epitope–tagged")	sft
HA	hyaluronic acid	sft
HAT	hypoxanthine, aminopterin, and thymidine	sft
HAT	histone acetyltransferase	sft
H^+ -ATPase	H^+ -dependent adenosine triphosphatase	sft
HAU	hemagglutination unit	sft
HAV	hepatitis A virus	sft
Hb	hemoglobin (deoxygenated)	sft
HbCO	carbon monoxide hemoglobin; carboxyhemoglobin	sft
HB-EGF	heparin-binding EGF-like growth factor	sft
HbO ₂	oxyhemoglobin	sft
HbS	sickle-cell hemoglobin	sft
HBSS	Hanks' balanced salt solution	sft
HBV	hepatitis B virus	sft
HC	hemicholinium	sft
hCG	human chorionic gonadotropin	sft
HCMV	human cytomegalovirus	sft
hcp	hexagonal close-packed	sft
HDAC	histone deacetylase	sft
HDF	Hubble Deep Field (HST image)	sft
HDF	human diploid fibroblast	sft
HDL	high-density lipoprotein	sft
HDPE	high-density polyethylene	sft
H&E	hematoxylin and eosin (a stain)	sft
HEK-293, HEK293, HEK 293	human embryonic kidney–293 (cells)	sft as "human embryonic kidney (HEK) 293 cells"
HeLa cell	(cell line)	OK as is
Hepes	4-(2-hydroxyethyl)-1-piperazineethane-sulfonic acid	OK as is
HEV	high endothelial venule	sft
HexNAc	<i>N</i> -acetylhexosamine	sft
HHH	hypothalamic hypophysiotropic hormone	sft
5-HIAA	5-hydroxyindoleacetic acid	sft
HIDE	human insulin-degrading enzyme	sft
HIF	hypoxia-inducible factor	sft
HIOMT	hydroxyindole- <i>O</i> -methyltransferase	sft
HIV	human immunodeficiency virus	OK as is
HIV-1, -2	human immunodeficiency virus–1, –2	OK as is

HLA	human lymphocyte antigen, human leukocyte antigen (one of a class of histocompatibility antigens) [Author preference]	sft
HLA-A, HLA-B	(classes of HLA)	OK as is if HLA has been previously defined
HMBC	heteronuclear multiple-bond correlation	sft
hMG	human menopausal gonadotropin	sft
HMG-CoA	3-hydroxy-3-methylglutaryl coenzyme A	sft
HMG box	(high-mobility group, family of transcription factors)	OK as is
HMQC	heteronuclear multiple-quantum coherence	sft
H-MSV	Harvey murine sarcoma virus	sft
HMTase	histone methyltransferase	sft
hnRNA	heterogeneous nuclear RNA	sft
hnRNP	heterogeneous nuclear RNP	sft
HOAc	acetic acid	Spell out
HOMO	highest occupied molecular orbital	sft
HPBW	half-power beam width	sft
HPE	hydroxyperoxy prostaglandin E	sft
HPLC	high-performance liquid chromatography	sft
HP _r	histidine-containing protein	sft
HPRT	hypoxanthine-guanine phosphoribosyltransferase	sft
HPTLC	high-performance thin-layer chromatography	sft
HREM	high-resolution electron microscopy, microscope	sft
HRF	hemodynamic response function	sft
HRP	horseradish peroxidase	sft
HRSEM	high-resolution scanning electron microscope	sft
HRTEM	high-resolution transmission electron microscope	sft
HSCs	hematopoietic stem cells (usually plural)	sft
HSQC	heteronuclear single-quantum coherence	sft
HST	Hubble Space Telescope	sft
HSV	herpes simplex virus	sft
HSV-TK	herpes simplex virus–thymidine kinase	sft
5-HT	5-hydroxytryptamine (serotonin)	Introduce as “serotonin [5-hydroxytryptamine (5-HT)]”
HTLV	human T cell leukemia virus	sft
5-HTP	5-hydroxytryptophan	sft
HTSC	high-temperature superconductivity	sft
HUVECs	human umbilical cord endothelial cells (usually plural)	sft
HVA	homovanillic acid (4-hydroxy-3-methoxyphenylacetic acid)	sft
HV volume	void volume	sft (also <i>V_o</i>)
H-Y antigen	histocompatibility-Y antigen	sft

HWHM

half width at half maximum

sft

<i>I, i</i>	current	sft
<i>I_{Ca}</i>	calcium current	sft
<i>Ia</i>	(class II histocompatibility antigens)	OK as is
IAA	indole-3-acetic acid (don't use for iodoacetic acid; spell out)	sft
IAPP	islet amyloid polypeptide	sft
IBMX	3-isobutyl-1-methylxanthine	sft
IC	intracarotid	sft
IC ₅₀	median inhibitory concentration (the concentration of a substance required to inhibit the activity of another substance by 50%)	sft
ICAM-1	intercellular adhesion molecule-1	sft
ICB	inner core boundary	sft
ICBM	International Conference on Brain Mapping (coordinate system)	sft
ICE	interleukin-1 β converting enzyme	sft
ICM	inner cell mass	sft
ICP-MS, ICPMS	inductively coupled plasma mass spectrometry	sft
ICP-OES	inductively coupled plasma optical emission spectrometry	sft
icv	intracerebroventricularly (after a dosage value only)	sft (but always spell out "intracerebroventricular")
ID	inside diameter	sft
ID ₅₀	median infective dose	sft
ID ₅₀	median inhibitory dose	sft
IDDM	insulin-dependent diabetes mellitus	sft
IDP	inosine 5'-diphosphate	sft (5'- may be omitted)
IdU	5'-iododeoxyuridine	sft
IEF	isoelectric focusing	sft
IELs	intraepithelial lymphocytes (usually plural)	sft
IEM	immunoelectron microscopy	sft
IETS	inelastic electron tunneling spectroscopy	sft
IFN	interferon	sft
IFN- α	interferon- α [define as "interferon- α (IFN- α)," not "interferon (IFN)- α "]	sft
IFN- β	interferon- β	sft
IFN- γ	interferon- γ (preferred to γ -IFN)	sft
IGF-1	insulin-like growth factor 1	sft
IgG	immunoglobulin G (IgG2a, IgG2b, etc.; full-size numbers)	sft
IgM	immunoglobulin M	sft
IGRF	International Geomagnetic Reference Field	sft

I κ B	inhibitor of nuclear factor κ B	Ensure that meaning is clear from context
IKK	I κ B kinase (subunits: IKK α , IKK β)	sft
IL-1	interleukin-1, also IL-2, IL-3, IL-4, IL-5 [define as “interleukin-1 (IL-1),” not “interleukin (IL)-1”]	sft
IL-1 α , IL-1 β	interleukin-1 α , interleukin-1 β	sft
ILK	integrin-linked kinase	sft
IMAC	immobilized metal affinity chromatography	sft
IMP	inosine 5'-monophosphate	sft (5'- may be omitted)
INAA	intraepithelial lymphocytes (usually plural)	sft
iNOS	inducible nitric oxide synthase	sft
INS	inelastic neutron scattering	sft
InSAR	interferometric synthetic aperture radar	sft
INT	paraiondonitrotetrazolium violet (for enzyme measurement)	sft
ip	intraperitoneally (after a dosage value only)	sft (but always spell out “intraperitoneal”)
IP	induced protein	sft
IP	ionization potential	sft
IP ₃	inositol 1,4,5-trisphosphate	sft (note tris)
IP ₄	inositol 1,4,5,6-tetrakisphosphate	sft
IP ₅	inositol 1,3,4,5,6-pentakisphosphate	sft
IP ₆	inositol hexakisphosphate	sft
IPCs	insulin-producing cells (usually plural)	sft
IPG	immobilized pH gradient (electrophoresis)	sft
IPG	inositolphosphoglycan	sft
IPIS	ion pair imaging spectroscopy	sft
IPSC	inhibitory postsynaptic current	sft
IPSP	inhibitory postsynaptic potential	sft
IPTG	isopropyl- β -D-thiogalactopyranoside	sft
IQ	intelligence quotient	sft
IQR	interquartile range	sft
IR	infrared	sft
IRAK	interleukin-1 receptor-associated kinase	sft
IRD	ice-rafted debris, ice-rafted detritus	sft (follow author)
IRES	internal ribosomal entry site	sft
IRF	interferon regulatory factor	sft
IRM	interference reflection microscopy	sft
IRM	isothermal remanent magnetization	sft
IRP	international reference preparation	sft
IRPTS	infrared-induced population transfer spectroscopy	sft
ISCCP	International Satellite Cloud Climatology Project	sft

ISRE	interferon-stimulated response element	sft
ISS	ion-scattering spectroscopy	sft
ITAM	immunoreceptor tyrosine-based activation motif	sft
ITC	isothermal titration calorimetry	sft
ITCZ	intertropical convergence zone	sft
ITIM	immunoreceptor tyrosine-based inhibitory motif	sft
ITO	indium tin oxide	sft
ITP	inosine 5'-triphosphate	sft (5'- may be omitted)
ITR	inverted terminal repeat	sft
ITT	insulin tolerance test	sft
iv	intravenously (after a dosage value only)	sft (but always spell out “intravenous”)
<i>I-V</i>	current-voltage relation	sft
IXS	inelastic x-ray scattering	sft

J	rotational constant (in spectroscopy)	sft
J	coupling constant in NMR	sft
J _H	heavy-chain joining region	sft
JAK	Janus kinase	sft
JFD	joint frequency distribution	sft
JNK	c-Jun N-terminal kinase (same as SAPK)	sft

k_B	Boltzmann constant	sft (change kT to $k_B T$ and query)
κ (kappa)	one of two possible light chains of an immunoglobulin (the other is λ , lambda)	sft
$[K^+]$, $[K^+]_o$	concentration and extracellular concentration of potassium (sub lc oh)	sft
K	equilibrium constant	sft
K_a	acid constant	sft
K_d	dissociation constant (binding affinity)	sft
K_d	distribution constant in chromatography	sft
K_e	equilibrium constant	sft
K_i	inhibition constant	sft
K_{ii}	inhibition constant determined from intercept rather than from abscissa	sft
K_m	Michaelis constant	sft (affinity for substrate; sometimes K_s)
ka	thousand years (ago)	sft
	[Commonly used to represent age. “Ago” is often implied but may be written out (“xx thousand years ago”) when context is needed (follow authors).]	
KA	kainate or kainic acid	sft
kb	kilobase	OK as is
KCN	potassium cyanide	OK as is
17-KGS	17-ketogenic steroid	sft
KI	Kovats indices (unbranched paraffin standards)	sft
KIR	killer cell immunoglobulin-like receptor	sft
KLH	keyhole limpet hemocyanin	sft
KNCS	potassium isothiocyanate	OK as is
KOAc	potassium acetate	sft
KRBGA	Krebs-Ringer-bicarbonate-glucose albumin	sft
KRG	Krebs-Ringer-glucose	sft
17KS	17-ketosteroid	sft
KSCN	potassium thiocyanate	OK as is
KSR-1	kinase suppressor of Ras-1	sft
K-T	Cretaceous-Tertiary (boundary)	sft
kyr	thousand years	sft

[Commonly used to represent duration/length of time. Phrases may also be written out (“xx thousand years”).]

<i>l</i> -	levorotatory (but IUPAC recommends using a minus sign instead)	OK as is (but see IUPAC-IUB)
L	indicates stereoisomerism (small caps)	OK as is
$L_{\odot}$	luminosity of the sun	sft
L-type calcium channel	(roman L)	OK as is
LAI	leaf area index	sft
LA-ICPMS	laser ablation inductively coupled plasma mass spectrometry	sft
LAK cell	lymphokine-activated killer cell	sft
L amino acid	stereochemically related to L-glyceraldehyde (not levorotatory)	OK as is, small cap L
λ (lambda)	one of two possible light chains of an immunoglobulin (other is κ , kappa)	sft
LANDSAT	satellite name	OK as is
LAT	linker of activated T cells	sft
LB, LB medium	Luria broth	OK as is
L cells	mouse fibroblasts	sft
L chain	light chain of Ig molecule	sft
LCM	leukocyte-conditioned medium	sft
LC-MS	liquid chromatography–mass spectrometry	sft
LCMV	lymphocytic choriomeningitis virus	sft
LCP	left circularly polarized (light)	sft
LCR	locus control region	sft
LD	lethal dose	sft
LD ₅₀	median lethal dose	sft
LDA	local density approximation	sft
LDFA	linear discriminant function analysis	sft
LDH	lactate dehydrogenase	sft
LDL	low-density lipoprotein	sft
L-dopa	levodopa	OK as is (small cap L)
LDOS	local density of states	sft
LED	light-emitting diode	sft
LEED	low-energy electron diffraction	sft
LEEM	low-energy electron microscopy	sft
LEFM	linear elastic fracture mechanics	sft
LET	linear energy transfer	sft
LETS	large external transformation-sensitive protein	sft
LFA-1, -2, -3	leukocyte function-associated antigen–1, –2, –3	OK as is
LFM	lateral force microscopy	sft
LFP	local field potential	sft
LGM	Last Glacial Maximum (initial caps)	sft
LGN	lateral geniculate nucleus	sft
LH	luteinizing hormone	sft

LH-RH, LHRH	luteinizing hormone–releasing hormone; also called gonadotropin-releasing hormone (GnRH)	sft
LIDAR	light detection and ranging	sft
LIF	leukemia inhibitory factor	sft
LIF	laser-induced fluorescence	sft
LIM domain	<i>Lin-11 Isl-1 Mec-3</i>	sft
LINEs	long interspersed nuclear elements (usually plural)	sft
LMB	leptomycin B	sft
LMM	light meromyosin	sft
LMPCR	ligation-mediated polymerase chain reaction	sft
$\mu\text{L}^2\text{MS}$	microprobe laser desorption laser ionization mass spectrometry	sft
ln	natural logarithm (to base <i>e</i>)	OK as is
lncRNA	long noncoding RNA	sft
LNMCs	lymph node mononuclear cells (usually plural)	sft
lod score	logarithm of the odds ratio for linkage	sft
log	logarithm (to base 10)	OK as is
LOH	loss of heterozygosity	sft
LPA	lysophosphatidic acid	sft
LPC	lysophosphatidylcholine	sft
LPG	liquefied petroleum gas	sft
LPL	lipoprotein lipase	sft
LPM	lateral plate mesoderm	sft
LPO	lattice-preferred orientation	sft
LPP	lipid phosphate phosphohydrolase	sft
LPS	lipopolysaccharide	sft
LRB	lissamine rhodamine B	sft
LREE	light rare earth element	sft
LRET	luminescence resonance energy transfer	sft
LRM	laser Raman microprobe	sft
LRR	leucine-rich repeat	sft
LSD	D-lysergic acid diethylamide (small cap D)	sft
LSD	least significant difference or digit	sft
LSDIC	laser scanning differential interference contrast	sft
LTC	long-term culture	sft
LTD	long-term depression	sft
LTF	long-term facilitation	sft
LTP	long-term potentiation	sft
LTR	long terminal repeat	sft
LUMO	lowest unoccupied molecular orbital	sft
LXR	liver X receptor (a nuclear receptor)	sft
Ly2	(an antigen)	OK as is
Lyn	(Src-like tyrosine kinase)	OK as is

M	mitosis (stage of cell cycle)	Ensure that meaning is clear from context
M	metal (spectroscopy)	OK as is
$M_{\odot}$	mass of the Sun, or solar mass	sft
m_b	body wave magnitude (earthquakes) (l.c. <i>m</i>)	sft
m_e	electron mass	sft
M_r	relative molecular mass	sft
M_n	number-average molecular weight	
M_w	weight-average molecular weight [Molecular weight, M_n , and M_w may be expressed with or without units (follow author). There are two accepted definitions for “molecular weight”: (i) a dimensionless molecular weight ratio (from IUPAC), which is synonymous with relative molecular mass, and (ii) a quantity expressed in units of g/mol (from the polymer community). Units of daltons or u (unified atomic mass units) are used for molecular mass.]	
M_s	surface wave magnitude (earthquakes)	sft
M_w	moment magnitude (earthquakes)	sft
β_2M	β_2 -microglobulin	sft
M-199	Medium 199 (Gibco)	sft
Ma	million years (ago)	sft
	[Commonly used to represent age. “Ago” is often implied but may be written out (“xx million years ago”) when context is needed (follow authors).]	
mAb	monoclonal antibody (plural: mAbs)	sft
MACS	magnetic cell sorter OR magnetic-activated cell sorting	sft
MAD	multiwavelength anomalous dispersion	sft
MADS, MADS box	(family of transcription factors: MCM1, AGAMOUS, DEFICIENS, and serum response factor)	OK as is
MAGUK	membrane-associated guanylate kinase	sft
MALDI	matrix-assisted laser desorption/ionization (mass spectrometry)	sft
MALDI-TOF	matrix-assisted laser desorption/ionization–time-of-flight (mass spectrometry)	sft
MALS	multiangle light scattering	sft
MALT	mucosa-associated lymphatic tissue	sft
MALToma	(change to MALT lymphoma)	
MAMP	microbe-associated molecular pattern	sft
Man	mannose or mannosidase	sft
ManNAc	<i>N</i> -acetylmannosamine	sft

MANOVA	multivariate analysis of variance	sft
MAO	monoamine oxidase	sft
MAPK	mitogen-activated protein kinase (same as ERK)	sft
MAPKAPK	MAPK-activated protein kinase	sft
MAPKK, MAP2K	MAPK kinase	sft
MAPKKK, MAP3K	MAPK kinase kinase (= MKK)	sft
MARK	microtubule affinity-regulating kinase	sft
MAS NMR	magic angle spinning NMR	sft
Mb	megabase	OK as is
Mb	myoglobin (deoxygenated)	sft (may be modified as HbCO and HbO ₂)
Mbp	mega-base pairs (treat as unit)	sft
MBP	myelin basic protein; maltose-binding protein (query author if necessary)	sft
MBE	molecular beam epitaxy, epitaxial	sft
MC-ICPMS	multicollector inductively coupled plasma mass spectrometry	sft
MCM protein	minichromosome maintenance protein	sft
MCMC	Markov chain Monte Carlo	sft
MCMM	Monte Carlo multiple minimum	sft
MCR	methyl-coenzyme M reductase	sft
M-CSF	macrophage colony-stimulating factor	sft
MDC	macrophage-derived chemokine	sft
MDCK cells	Madin-Darby canine kidney cells	sft
MDMA	(±)3,4-methylenedioxymethamphetamine ("ecstasy")	sft
MDS	multidimensional scaling	sft
MDTF	<i>Mus dunni</i> tail fibroblast (cell line)	sft
Me	methyl	sft
MeBMT	<i>N</i> -methyl-butemylthreonine	sft
MeCN	acetonitrile	sft
MECs	medullary epithelial cells (usually plural)	
MEFs	mouse embryonic fibroblasts (usually plural)	sft
MEF2	myocyte enhancer factor 2 (a family of transcription factors)	sft
MEG	magnetoencephalography, magnetoencephalogram	sft
MEK	MAPK kinase (originally "MAPK-ERK kinase," per, but rarely seen that way)	sft
MEM	minimum essential medium	sft (Eagle's or Dulbecco's)
MeOH	methanol	sft
MEPC	miniature end-plate current	sft
MEPP	miniature end-plate potential	sft

-mer	(do not use; change 40-mer to 40-nucleotide oligomer)	
MES	2-(<i>N</i> -morpholino)ethanesulfonic acid (a buffer)	OK as is
MET	metanephrine	sft
Met-enkephalin	methionyl enkephalin	OK as is
MetHB	methemoglobin	sft
MFM	magnetic force microscope, microscopy	sft
MG132	<i>N</i> -carbobenzyloxy-L-leucyl-L-leucyl-L-leucinal (a proteasoml inhibitor)	sft
mGluR	metabotropic glutamate receptor	sft
MHA	16-mercaptohexadecanoic acid	sft
MHA-HA	mixed hemadsorption–hybrid antibody test (en dash when spelled out)	sft
MHC	major histocompatibility complex	sft
MHCK	myosin heavy chain kinase	sft
MHD	magnetohydrodynamic	sft
MHPG	3-methoxy-4-hydroxyphenylglycol	sft
MIBK	methyl isobutyl ketone	sft
MIC	minimum inhibitory concentration	sft
MICA, MICB	(stress-inducible antigens)	OK as is
MIDAS	metal ion–dependent adhesion site	sft
MIF	migration inhibition factor	sft
MIF	melanocyte release inhibiting factor	sft
MIP	macrophage inflammatory protein	sft
MIPS	Munich Information Center for Protein Sequences	sft
MIR	main immunogenic region	sft
MIR	multiple isomorphic replacement (x-ray crystallography)	sft
MIRAS	multiple isomorphous replacement with anomalous scattering	sft
miRNA	microRNA	sft
MIS	marine isotope stage (e.g., MIS 12)	sft
MIS	Müllerian inhibiting substance	sft
MIX	3-isobutyl-1-methylxanthine	sft
MKK	MAPK kinase kinase	sft
MKP	MAPK phosphatase	sft
ML	monolayer (treat as unit)	sft
ML	maximum likelihood	sft
MLC	mixed lymphocyte culture	sft
MLC	myosin light chain	sft
MLCK	myosin light chain kinase	sft
MLD	minimum lethal dose	sft
MLR	mixed leukocyte reaction	sft
MMA	monomethylamine	sft
mmp	mixed melting point	sft

MMP	matrix metalloproteinase	sft
MMS	methyl methanesulfonate	sft
MNSs	anti-M, anti-N, anti-S, anti-s blood group	sft
MO	molecular orbital	sft
MOCVD	metal-organic chemical vapor deposition	sft
MOI	multiplicity of infection	sft
MoMLV	Moloney murine leukemia virus	sft
Mops	3-(<i>N</i> -morpholino)propanesulfonic acid	OK as is
MORB	mid-ocean ridge basalt	sft
MOSFET	metal oxide semiconductor field-effect transistor	sft
MOT	magneto-optical trap	sft
mp	melting point	sft
MPD	mean population doubling	sft
MPF	maturation promoting factor	sft
MPLSM	multiphoton laser scanning microscope, microscopy	
MPN-PCR	most-probable-number PCR	sft
MPR	mannose 6-phosphate receptor	sft
MPTP	1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine	sft
MR	magnetoresistance	sft
MRAM	magnetoresistive random access memory	sft
MRCA	most recent common ancestor	sft
MRI	magnetic resonance imaging	sft
mRNA	messenger RNA	Define only in abstracts
MS	mass spectrometry (usually follows the name of the technique)	sft
MSA	methane sulfonic acid	sft
MSCs	mesenchymal stem cells (usually plural)	sft
MSH	melanocyte-stimulating hormone	sft
MSH-IF	melanocyte-stimulating hormone inhibitory factor	sft
MSK	mitogen- and stress-activated protein kinase	sft
MSL	mean sea level	sft
MS/MS	tandem mass spectrometry	sft
MSV	murine sarcoma virus	sft
MSV	Moloney sarcoma virus	sft
MSWD	mean square weighted deviation	sft
mtDNA	mitochondrial DNA	sft
MTD	magnetic tunnel junction	sft
mTOR	mammalian target of rapamycin, mechanistic target of rapamycin (check with author)	sft
MTV	mammary tumor virus	sft
MuLV	murine leukemia virus	sft
MurNAc	<i>N</i> -acetylmuramic acid	sft
MW	molecular weight (allow with or without units; see M_w , M_n)	sft

M $\square$ WNT	multiwalled carbon nanotube (preferred over MWCNT)	sft
Myr	million years	sft
	[Commonly used to represent duration/length of time. Phrases may also be written out (“xx million years”).]	
<i>m/z</i>	mass/charge ratio	sft

N	normal (as in concentration)	OK as is
N-terminal, N terminus	amino-terminal, amino terminus	Spell out in abstract; otherwise OK as is
<i>N, n</i>	number of samples (italic)	OK as is
N-type calcium channel	(roman N)	OK as is
NA	noradrenaline	sft
NA	numerical aperture	sft
N.A., NA	not applicable	sft (query author to be sure)
NAA	naphthaleneacetic acid	sft
NAA	neutron activation analysis	sft
NAD	nitric acid dihydrate	sft
NAD	nicotinamide adenine dinucleotide	sft
NAD ⁺	nicotinamide adenine dinucleotide (oxidized form)	sft
NADH	reduced form of NAD ⁺	sft
NADP ⁺	nicotinamide adenine dinucleotide phosphate	sft
NADPH	reduced form of NADP ⁺	sft
NADW	North Atlantic Deep Water (initial caps)	sft
Na _E	exchangeable sodium	sft
Na ₂ HPO ₄ , NaH ₂ PO ₄	sodium phosphate buffer	Spell out
Na ⁺ , K ⁺ - ATPase	Na ⁺ - and K ⁺ -dependent ATPase (no space after comma)	sft
NANA	<i>N</i> -acetylneuraminic acid (same as NeuNAc)	sft
NAO	North Atlantic Oscillation (initial caps)	sft
NAT	nitric acid trihydrate	sft
NBQX	2,3-dioxo-6-nitro-7-sulfamoylbenzo[<i>f</i>]quinoxaline 6-nitro-7-sulfamoylbenzo[<i>f</i>]quinoxaline-2,3-dione (an NMDA antagonist)	sft (both forms are in use and apparently have the same meaning; query author)
NB-tropic	virus that grows well on NIH or Balb/c cells	sft
NBT	nitro blue tetrazolium	sft
NCAM	neural cell adhesion molecule	sft
NCBI	National Center for Biotechnology Information (database)	sft
NCIs	neuronal cytoplasmic inclusions	sft
ncRNA	noncoding RNA	sft
NCS	noncrystallographic symmetry	sft
ND, nd, n.d.	not determined, defined, done	sft (query author to be sure)
NDOS	normalized density of states	sft
NDVI	normalized difference vegetation index	sft
NE	norepinephrine	sft

NEM	<i>N</i> -ethylmaleimide	sft
NER	nucleotide excision repair	sft
NEt	ethylamine	sft
NeuNAc	<i>N</i> -acetylneuraminic acid (NANA)	sft
NEXAFS	near-edge x-ray absorption fine structure	sft
NF	nuclear factor	sft
NFAT	nuclear factor of activated T cells (a transcription factor)	Ensure that meaning is clear from context
NF-κB	nuclear factor κB (a transcription factor)	sft
NGF	nerve growth factor	sft
β-NGF	β-subunit of NGF	sft
NGS	normal goat serum	sft
NHEJ	nonhomologous end joining	sft
NHS	<i>N</i> -hydroxysuccinimide	sft
NIH 3T3	cell line	OK as is
NIK	NF-κB-inducing kinase (= MAPKKK)	sft
NK cells	natural killer cells	sft
N-linked	nitrogen-linked (roman N)	OK as is (use italic <i>N</i> in chemical names only)
NLS	nuclear localization signal	sft
NMD	nonsense-mediated mRNA decay	sft
NMDA	<i>N</i> -methyl-D-aspartate	sft
NMDG	<i>N</i> -methyl-D-glucamine	sft
NMET	normetanephrine	sft
NMN	nicotinamide mononucleotide	sft
NMNAT	nicotinamide mononucleotide adenylyl transferase	sft
NMR	nuclear magnetic resonance	sft
NOD mice	nonobese diabetic mice	sft
NOE	nuclear Overhauser effect	sft
NOESY	nuclear Overhauser effect spectroscopy	sft
NOS	nitric oxide synthase	sft
NPH	neutral protamine Hagedorn	sft
NP-40	(a detergent)	OK as is
NPRS	nonparametric rate smoothing	sft
NPY	neuropeptide Y	sft
NQR	nuclear quadrupole resonance	sft
NR	neutralization-reionization (mass spectrometry)	sft
NRIXS	nuclear resonant inelastic x-ray scattering	sft
NRM	natural remanent magnetization	sft
n.s.	not significant (statistics)	sft
NSF	<i>N</i> -ethylmaleimide-sensitive factor	sft
NSOM	near-field scanning optical microscopy	sft
nt	nucleotide (as unit after a number)	sft (but spell out when not used as

		unit, e.g., “nucleotides 22 to 37”)
NT ₅₀	end point of 50% reduction of virus expression	sft
NTA	nitrilotriacetic acid	sft
NTP	nucleotide triphosphate, nucleoside triphosphate	sft (query if undefined)
NTR	nontranslated region	sft (query author: change to UTR?)
NVP	naphthyl vinyl pyridyl or pyridine	sft

OAc	acetate	sft
OB stars	(O and B spectral class stars, the most massive stars in the main sequence)	Ensure that meaning is clear from context
OCS	carbonyl sulfide	sft
OD	outer diameter	sft
OD	optical density—use OD only for reduced light (or UV) transmission through turbid suspension; otherwise, change to <i>A</i> , absorbance	sft
ODA	octadecylamine	sft
ODB	<i>o</i> -dichlorobenzene	sft
ODMR	optically detected magnetic resonance	sft
ODNs	oligodeoxynucleotides (usually plural)	sft
ODP	Ocean Drilling Program (initial caps, usually modifies site or core)	sft
ODS	octadecylsilane	sft
ODT	1-octadecanethiol	sft
OES	optical emission spectrometry	sft
OFAGE	orthogonal field alternation gel electrophoresis	sft
11-OHCS	11-hydroxycorticosteroid	sft
17-OHCS	17-hydroxycorticosteroid	sft
6-OHDA	6-hydroxydopamine	sft
OLA	optimally localized averaging	sft
oligo(dT)	oligothymidilate	OK as is
OLS	ordinary least squares (statistical method)	sft
OMP	orotidine 5'-monophosphate	sft (5'- may be omitted)
OMZ	oxygen minimum zone	sft
OPA	optical parametric amplifier	sft
OPR	ortho-to-para ratio (astrophysics)	sft
ORC	origin recognition complex	sft
ORF	open reading frame	sft
ORTEP	Oak Ridge thermal ellipsoid plot	sft
OVA	ovalbumin	sft

<i>P</i>	pressure	sft
<i>P</i>	probability (of significant difference)	sft
<i>P</i> ₅₀	partial pressure of O ₂ at which hemoglobin is half-saturated	sft
<i>P</i> _a	arterial pressure	sft
<i>P</i> _A	alveolar pressure (small cap A)	sft
<i>P</i> _i	inorganic phosphate	sft
P-region	pore-forming region (of a cell membrane)	sft
P-type calcium channel	(roman P)	OK as is
<i>P</i> wave (also <i>PP</i> , <i>PPP</i>)	compressional wave (earthquake); specific waves may have numbers within, e.g., <i>P</i> 410 <i>P</i>	OK as is
p38	(a mammalian homolog of high-osmolarity glycerol response kinase, a MAPK; also known as SAPK)	OK as is
p42 ^{MAPK}	(a 42-kD MAPK; same as ERK2)	OK as is
p44 ^{MAPK}	(a 44-kD MAPK; same as ERK1)	OK as is
PABP	poly(A)-binding protein	sft
PAC	photoacoustic calorimetry	sft
PAC	perturbed angular correlation spectroscopy	sft
PACAP	pituitary adenylate cyclase activating polypeptide	sft
<i>P</i> _A CO ₂ , <i>P</i> ACO ₂	alveolar <i>P</i> CO ₂	Change to “alveolar <i>P</i> CO ₂ ”
PAF	platelet activating factor	sft
PAG	periaqueductal gray (brain region)	sft
PAGE	polyacrylamide gel electrophoresis	sft
PAHs	polycyclic aromatic hydrocarbons (usually plural)	sft
PAK	p21-activated protein kinase	sft
PALA	<i>N</i> -(phosphonoacetyl)-L-aspartate (small cap L)	sft
PAM	pyridine-2-aldoxime methiodide	sft
PAM	1) point accepted mutation 2) protospacer adjacent motif	sft
PAMP	pathogen-associated molecular pattern	sft
PAN	puromycin aminonucleoside	sft
PAN	peroxyacetyl nitrate	sft
PAP	peroxidase-antiperoxidase	sft
PAPS	3'-phosphoadenosine 5'-phosphosulfate	sft
PARP	poly(ADP-ribose) polymerase	sft
PARPi	poly(ADP-ribose) polymerase inhibitor	
PAS	periodic acid–Schiff	sft
PAS	Per-Arnt-Sim (usually followed by “domain”; transcription factor family)	sft
PBG	photonic bandgap	sft
PBLs	peripheral blood lymphocytes, peripheral blood leukocytes (usually plural) [Author preference]	sft
PBMCs	peripheral blood mononuclear cells (usually plural)	sft
PBS	phosphate-buffered saline	sft

PC	phosphatidylcholine	sft
PC	personal computer	OK as is
PCA	perchloric acid	sft
PCA	principal components analysis (not “component”)	sft
<i>p</i> -CA	<i>p</i> -chloroamphetamine	sft
pCa	$-\log[\text{Ca}^{2+}]$	OK as is
PCB	polychlorinated biphenyl	sft
PcG	Polycomb group (proteins)	sft
PCNA	proliferating cell nuclear antigen	sft
<i>P</i> CO ₂	partial pressure of CO ₂ (small cap CO)	sft
PCP	phencyclidine	sft
PCR	polymerase chain reaction	sft
PCV	packed cell volume	sft
PCV	percent coefficient of variation	sft
<i>Pd</i>	diastolic pressure	sft
PDB	Pee Dee belemnite (standard for core sampling)	sft
PDB	Protein Data Bank (Brookhaven National Laboratory)	sft
PDF	pigment-dispersing factor	sft
PDGF	platelet-derived growth factor	sft
PDGFR	platelet-derived growth factor receptor	sft
PDH	pyruvate dehydrogenase	sft
PDI	protein disulfide isomerase	sft
PDMS	polydimethylsiloxane, poly(dimethylsiloxane) [AU preference]	sft
pDRM	postdepositional remanent magnetization	sft
PDZ domain	(a protein interaction domain; name derives from proteins that contain the domain: PSD-95, Dlg, ZO1)	OK as is
PE	phycoerythrin	sft
PE	phosphatidylethanolamine	sft
PEC	photoelectric cell	sft
PECs	peritoneal exudate cells (usually plural)	sft
PECAM	platelet endothelial cell adhesion molecule	sft
PECT	phosphoethanolamine cytidyltransferase	sft
PECVD	plasma-enhanced chemical vapor deposition	sft
PEDF	pigment epithelium–derived factor	sft
PEEM	photoelectron emission microscope, microscopy	sft
PEG	polyethylene glycol, poly(ethylene glycol) [AU preference]	sft
PEG-800	polyethylene glycol, molecular weight 800 (similarly for other molecular weights)	sft
PEP	phosphoenolpyruvate	sft
PES	potential energy surface	sft
PET	positron emission tomography	sft
PET	photosynthetic electron transport	sft
PET	polyethylene terephthalate	sft

PFGE	pulsed-field gel electrophoresis	sft
PFOR	pyruvate:ferredoxin oxidoreductase	sft
PFU	plaque-forming units	sft
	(“s” generally not required for plural, but if both individual and plural are mentioned, use PFU/PFUs)	
PG	Prostaglandin	sft
PGE	prostaglandin E (variants: PGE ₂ , PGE _{2α})	sft
PGI ₂	prostacyclin	sft
PGO waves	ponto-geniculo-occipital waves	sft
pH	−log[H ⁺], negative logarithm of the hydrogen ion concentration	OK as is
Ph	phenyl	sft
Ph ¹	Philadelphia chromosome	sft (do not use Ph ⁺ for Philadelphia 1-positive)
PH domain	pleckstrin homology domain	sft
PHA	phytohemagglutinin	sft
PHGPx	phospholipid hydroperoxide glutathione peroxidase	sft
pI	isoelectric point	sft
PI	propidium iodide	sft
PI	phosphatidylinositol	sft
PI 3-kinase	phosphatidylinositol 3-kinase (in contexts where PtdIns is defined this way, PI 3-kinase should be defined as phosphoinositide 3-kinase)	sft
PI3K	(shorter form for PI 3-kinase if used frequently)	sft
PIC	protease inhibitor cocktail	sft
PIF	prolactin inhibitory factor	sft
PIMC	path-integral Monte Carlo (simulation)	sft
PI(4,5)P ₂ , PIP ₂	phosphatidylinositol 4,5-bisphosphate	sft (note bis)
PI(3,4,5)P ₃ , PIP ₃	phosphatidylinositol 3,4,5-trisphosphate	sft (note tris)
Pipes	1,4-piperazinediethanesulfonic acid (buffer)	OK as is
piRNA	Piwi-interacting RNA	sft
PITC	phenyl isothiocyanate	OK as is
PIU	polymerase-inducing units	sft
p <i>K</i>	[Query author to specify which <i>K</i> if not already specified]	sft
p <i>K</i> _a	Add “(where <i>K</i> _a is the acid dissociation constant)” if possible to do so	sft
PKA	cAMP-dependent protein kinase	sft
PKB	protein kinase B (= Akt)	sft
PKC	protein kinase C	sft
PKG	cGMP-dependent protein kinase	sft
PKR	RNA-activated protein kinase	sft
PKU	phenylketonuria	sft
PLA ₂	phospholipase A ₂	sft

PLAC	plant artificial chromosome	sft
PLC	phospholipase C	sft
PLC- β , PLC- γ	phospholipase C- β , C- γ (specific isoforms of PLC; use en dash when spelling out)	sft
PLD	phospholipase D	sft
PLD	pulsed laser deposition	sft
PLK	Polo-like kinase	sft
PMA	phorbol 12-myristate 13-acetate	sft
PMMA	poly(methyl methacrylate), polymethyl methacrylate [AU preference]	sft
PMR	proton magnetic resonance	sft
PMSF	phenylmethanesulfonyl fluoride	sft
PMT	photomultiplier tube	sft
PNA	peanut agglutinin	sft
PND	postnatal day (more commonly referred to with just “P” e.g., “postnatal day 3 (P3)”)	sft
PNGase	peptide <i>N</i> -glycosidase	sft
PNMT	phenylethanolamine- <i>N</i> -methyltransferase	sft
PNS	peripheral nervous system	sft
POC	particulate organic carbon	sft
PO_2	partial pressure of oxygen; oxygen tension (partial pressure of oxygen within bloodstream)	sft
Pol	polymerase, as in Pol II, Pol III (cap P)	sft
poly(A)	polyadenylate (synthetic polynucleotide composed of adenylate residues)	sft
poly(A) ⁺	[not poly(A ⁺)] polyadenylated	sft
poly(X,Y)	random copolymer of X and Y, where X and Y are nucleotides A, C, T, G, or U	sft
POM	polarized optical microscopy	sft
POMC	proopiomelanocortin	sft
POPE	palmitoyloleoylphosphatidylethanolamine	sft
POPOP	1,4-bis[2-(5-phenyloxazole)]-benzene (a reagent)	sft
PPAR	peroxisome proliferator-activated receptor (family of nuclear receptors: PPAR α , PPAR β , PPAR γ/δ)	sft
PP1, PP2B	protein phosphatase 1, protein phosphatase 2B	sft
PPD	purified protein derivative	sft
PP _i	inorganic pyrophosphate	sft
PPO	2,5-diphenyloxazole (for liquid scintillation)	sft
PPV	poly(<i>p</i> -phenylenevinylene)	sft
Ppy	polypyrrole	sft
PQ	plastoquinone	sft
<i>Pr</i>	Prandtl number	sft
pre-B cells, pre-T cells	precursor B cells, precursor T cells (use hyphen, not en dash)	sft
PREM	Preliminary Reference Earth Model	sft
pre-mRNA	precursor mRNA	sft

PRF	prolactin-releasing factor	sft
PRL	prolactin	sft
pro-B cells, pro-T cells	a B cell progenitor stage that comes before pre-B cell (use hyphen, not en dash)	OK as is
PROM	programmable read-only memory	sft
PrP, PrP ^C , PrP ^{Sc}	prion protein PrP ^C : the normal cellular prion protein PrP ^{Sc} : the abnormal disease-causing isoform	sft
PRR	pattern recognition receptor	sft
<i>Ps</i>	systolic pressure	sft
PSA	prostate-specific antigen	sft
PSD	postsynaptic density	sft
PSD	post-source decay (spectral analysis)	sft
PSI, PSII	photosystem I, II	sft
PSK	phytosulfokine	sft
PSP	postsynaptic potential	sft
PS-PBMA	polystyrene-polybutylmethacrylate	sft
PSRE	proton spin relaxation editing	sft
PST	poststimulus time	sft
PSTH	poststimulus time histogram, peristimulus time histogram	sft (query author for desired form)
PSU	practical salinity unit	sft
P-T	Permian-Triassic (boundary)	sft
PTB	phosphotyrosine binding domain	sft
PTC	premature termination codon	sft
PtdIns	see PI 3-kinase; sometimes phosphatidylinositol 3- kinase	sft
PTE	6 M NaClO ₄ , 10 mM tris-acetate, and 1 mM EDTA (pH 8.0)	sft
PTGS	picosecond transient grating spectroscopy	sft
PTGS	posttranscriptional gene silencing	sft
PTH	phenylthiohydantoin	sft
PTK	protein tyrosine kinase	sft
PTP	protein tyrosine phosphatase	sft
PTTP	prothoracicotropic hormone	sft
PTU	propylthiouracil	sft
PTx	pertussis toxin	sft
PVC	polyvinyl chloride, poly(vinyl chloride) [AU preference]	sft
PVDF	polyvinylidene difluoride	sft
PVE	phenotypic variance explained (expressed as percent)	sft
PVP	polyvinyl pyrrolidone, poly(vinyl pyrrolidone) [AU preference]	sft
PXR	pregnane X receptor (a nuclear receptor)	sft
PXRD	powder x-ray diffraction	sft

Pyk2	(a nonreceptor tyrosine kinase)	OK as is
PZT	lead-zirconate-titanate (piezoelectric transducer)	sft

Q	ubiquinone	sft
Q	scattering vector	sft
$\dot{Q}$	flow rate	sft
$\dot{Q}$	gas production rate (astrophysics); use subscript for gas, e.g., $\dot{Q}_{\text{CO}}$	sft
Q_{10}	increase in rate of a chemical reaction for each 10°C increase in temperature	sft
QC	quantum cascade (laser)	sft
QCP	quantum critical point	sft
QDSL	quantum dot superlattice	sft
QED	quantum electrodynamics	sft
QNB	quinuclidonol benzoate	sft
QPC	quantum point contact	sft
QPCR	quantitative real-time fluorescence polymerase chain reaction	sft
qPCR	quantitative polymerase chain reaction	
QPT	quantum phase transition	sft
QTL	quantitative trait locus (QTLs, quantitative trait loci)	sft
Q-TOF	quadrupole orthogonal acceleration–time-of-flight (mass spectrometry)	sft
QWIP	quantum-well infrared photodetector	sft

r	correlation coefficient	sft
R	Rayleigh number	sft
R_E	Earth radius, radii (treat as unit)	sft
R_F	(sub cap f; all italic) retention characteristics (in TLC or paper chromatography), the ratio of the movement of the solute to that of the solvent front	sft
R_f	(sub lc f; all italic) retardation factor (distance the unknown has traveled relative to the solvent front)	sft
r_h	heliocentric distance, distance from the Sun	sft
R loop	three-stranded DNA-DNA-DNA hybrid	OK as is
r-process	rapid nucleosynthetic process (star formation)	Ensure that meaning is clear from context
R state	relaxed state of enzyme	sft
R_1 wave	(same as S wave)	OK as is
Rac	(a small GTPase)	OK as is
RACE	rapid amplification of cDNA ends	sft
R&D	research and development	OK as is
Raf	(a mitogen-activated protein kinase kinase kinase)	OK as is
RAM	random-access memory	sft
Ran, Rap, Ras	(types of small GTPase)	OK as is
RANTES	(a human chemokine: “regulated on activation, normal T cell–expressed and secreted”)	OK as is
RBCs	red blood cells (usually plural)	sft
RBP	ribose-binding protein	sft
RCP	right circularly polarized (light)	sft
RCS	regional curve standardization	sft
rDNA	recombinant DNA <i>or</i> ribosomal DNA (must be defined first time and used consistently throughout)	sft
RdRP	RNA-dependent RNA polymerase	sft
Re	Reynolds number	sft
REF	rat embryo fibroblast	sft
REM	rapid eye movement	sft
REMPI	resonance enhanced multiphoton ionization (preferred over “resonant” or “resonantly”)	sft
rf	radio frequency	sft
RF	releasing factor	sft
RF	rheumatoid factor	sft
RF	replicative form	sft
RF	restriction fragment (of DNA)	sft
RFLP	restriction fragment length polymorphism	sft
RFP	red fluorescent protein	sft
RGCs	retinal ganglion cells (usually plural)	sft
RGS domain	regulator of G protein signaling domain	sft
RHEED	reflection high-energy electron diffraction	sft
Rho	(a small GTPase)	OK as is

RIA	radioimmunoassay	sft
Rib	ribosyl, ribose	sft
RIF	radioimmunofluorescence	sft
RING, RING finger	(an adapter protein domain)	OK as is
RIPA	radioimmunoprecipitation assay	sft
RIPK	receptor-interacting protein kinase	sft
RISC	RNA-induced silencing complex	sft
RIXS	resonant inelastic x-ray scattering	sft
RLS	regularized least squares	sft
RM	red marrow cells	sft
RMS	root mean square	sft
RMSD	root mean square deviation	sft
RNA	ribonucleic acid	OK as is
RNAi	RNA interference	sft
RNAP	RNA polymerase	sft
RNase	ribonuclease: RNase A (pancreatic), RNase T (from plant), RNase H (from bacterial enzyme)	sft
RNA-seq	RNA sequencing	sft
RNP	ribonucleoprotein	sft
ROI	region of interest	sft
ROS	reactive oxygen species	sft
rpHPLC, RP-HPLC	reversed-phase HPLC	sft (spell out HPLC first time)
RPMI 1640	Roswell Park Memorial Institute medium	OK as is
RPW	relativistic plasma wave	sft
RQ	respiratory quotient	sft
rRNA	ribosomal RNA	sft
RSK	ribosomal protein S6 kinase	sft
RSV	Rous sarcoma virus	sft
RT	reverse transcriptase	sft
<i>RT</i>	Boltzmann gas constant $\times$ temperature	OK as is, if used together
RTD	resonant tunneling device	sft
rtDNA	recombinant DNA	sft
RTK	receptor tyrosine kinase	sft
RTOC	reaggregate thymic organ culture	sft
RT-PCR	reverse transcription polymerase chain reaction	sft
RU-486	(an abortifacient)	OK as is
RuBisCO	ribulose-1,5-bisphosphate carboxylase/oxygenase	sft
RVB	resonating valence bond	sft
RXR	retinoid X receptor (family of nuclear receptors)	sft

S phase	period of DNA replication in the cell cycle	Ensure that meaning is clear from context
<i>S</i> , as in 8 <i>S</i>	Svedberg unit (sedimentation coefficient)	OK as is
S band	(a band of radio frequencies, 1550 to 5200 MHz)	sft
s-process	slow nucleosynthetic process (star formation)	Ensure that meaning is clear from context
<i>S</i> wave (also <i>SS</i> , <i>SSS</i>)	shear wave (earthquake); specific waves may have numbers within, e.g., <i>S</i> 660 <i>S</i>	OK as is
ΔS	change in entropy (thermodynamics variable)	sft
<i>S</i> ₁ , <i>S</i> ₂ , <i>S</i> ₃	selected generations (genetics)	OK as is
S1P	sphingosine 1-phosphate	sft
S6K1	S6 kinase 1	sft
<i>S</i> _{20w} = 5.5	<i>S</i> = 5.5 at 20°C in water (Svedberg unit)	sft
S-20 fraction	fraction separated by centrifugation at 20,000 <i>g</i>	sft
SAED	selected-area electron diffraction	sft
SAGE	serial analysis of gene expression	sft
sALT	serum alanine aminotransferase	sft
SAM	self-assembled monolayer	sft
SANS	small-angle neutron scattering	sft
SAPK	stress-activated protein kinase (same as JNK)	sft
SAR	synthetic aperture radar	sft
SARS-CoV	severe acute respiratory syndrome coronavirus (not SARS-CoV-1)	sft
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2	sft
SAXR	small-angle x-ray reflection	sft
SAXS	small-angle x-ray scattering	sft
SBP	systolic blood pressure	sft
sc	subcutaneously (after a dosage value only)	sft (but always spell out “subcutaneous”)
SCCM	standard cubic centimeter per minute	sft
SCE	saturated calomel electrode	sft
SCF	stem cell factor	sft
SCF	SKP1/cullin/F-box protein [per , “SCFs consist of four subunits: a cullin, SKP1 (S-phase kinase-associated protein 1), a RING finger protein (RBX1/HRT1/ROC1), and an F-box protein”]	sft
SCFT	self-consistent field theory	sft
SCID	severe combined immunodeficient (mice); severe combined immunodeficiency disease	sft
SCN	suprachiasmatic nucleus	sft
SCN ⁻	thiocyanate (superscript minus sign)	sft
SCUBA		OK as is
SD	standard deviation	OK as is

SD	serologically defined	sft
SD ₅₀	50% suppression of mitogen-induced lymphoproliferation	sft
SDS	sodium dodecyl sulfate; used in polyacrylamide gel	OK as is
SDS-PAGE	SDS–polyacrylamide gel electrophoresis	sft PAGE; SDS OK as is
SDW	spin density wave	sft
SE	standard error	OK as is
SEC	size exclusion chromatography	sft
SECIS	selenocysteine insertion sequence	sft
SECS	single electron capacitance spectroscopy	sft
SELDI	surface-enhanced laser desorption/ionization (mass spectrometry)	sft
SEM	standard error of the mean	OK as is
SEM	scanning electron microscopy, microscope	sft
SEMPA	scanning electron microscopy with polarization analysis	sft
SEP	solar energetic particle	sft
SER	smooth endoplasmic reticulum	sft
SERM	selective estrogen receptor modulator	sft
SERS	surface-enhanced Raman scattering (spectroscopy)	sft
SGM	scanned-gate microscopy	sft
SGPE	starch-gel protein electrophoresis	sft
sgRNA	single guide RNA	sft
SH1 domain	Src homology 1 domain	sft
SH2 domain	Src homology 2 domain	sft
SH3 domain	Src homology 3 domain	sft
<i>SH</i> wave	horizontal shear wave (earthquake)	OK as is
<i>SV</i> wave	vertical shear wave (earthquake)	OK as is
SHE	standard hydrogen electrode	sft
SHIP	SH2-containing inositol polyphosphate 5-phosphatase	sft
SHP	small heterodimer partner (an orphan receptor)	sft
SHPM	scanning Hall probe microscope	sft
shRNA	short hairpin RNA	sft
shRNAi	short hairpin–mediated RNA interference	sft
SID	Smad interaction domain	sft
sIgA	secretory immunoglobulin A	sft
SIMS	secondary ion mass spectrometry	sft
SINEs	short interspersed nuclear elements (usually plural)	sft
SIR	single isomorphous replacement	sft
SIRAS	single isomorphous replacement plus anomalous scattering	sft
siRNA	small interfering RNA	sft

SIR/SAS	single isomorphous replacement and single anomalous scattering	Change to SIRAS
SIV	simian immunodeficiency virus	sft
SKIP	Ski oncogene interacting protein	sft
SLOT	scanning laser optical trapping	sft
SMAC	(usually appears as “SMAC/Diablo”): second mitochondria-derived activator of caspases	OK as is
SMAD	(a signal transducer protein family; members Smad1 to Smad8, initial cap only)	OK as is
SMARD	single-molecule analysis of replicated DNA	sft
SMC	smooth muscle cell	sft
SME	stalk-medium eminence tissue	sft
SMOW	standard mean ocean water	sft
<i>sn</i>	stereospecifically numbered (in chemical name)	OK as is
S _N 1, S _N 2	first-order, second-order nucleophilic displacement reaction	Ensure that meaning is clear from context
S/N ratio, SNR	signal-to-noise ratio	sft
SNAP	soluble NSF attachment protein	sft
SNARE	SNAP receptor	sft
SND	single-nucleotide discrepancy	sft
snoRNA	small nucleolar RNA	sft
snoRNP	small nucleolar ribonucleoprotein	sft
SNP	single-nucleotide polymorphism	sft
snRNA	small nuclear RNA	sft
snRNP	small nuclear ribonucleoprotein	sft
sRNA	small RNA	sft
SOD	superoxide dismutase	sft
SOI	Southern Oscillation Index (initial caps)	sft
SOS	(Son of Sevenless, a Ras guanine nucleotide–releasing factor)	OK as is
sp.	as in <i>Tridnaca</i> sp. (do not italicize sp.)	OK as is
SPC	sphingosylphosphoryl choline	sft
SPF	specific pathogen–free (en dash)	sft
SPH	smoothed particle hydrodynamics	sft
SPL	scanning probe lithography	sft
SPL	sound pressure level	sft
SPLEEM	spin-polarized low-energy electron microscopy	sft
SPM	scanning probe microscope	sft
SPO	shape-preferred orientation	sft
SPOT	Système pour l’Observation de la Terre	sft
SPR	surface plasmon resonance	sft
SP-STM	spin-polarized scanning tunneling microscope, microscopy	sft
SPT	serine palmitoyltransferase	sft
SQUID	superconducting quantum interference device	sft

SR	sarcoplasmic reticulum	sft
Src	(a nonreceptor tyrosine kinase)	OK as is
SRE	serum response element	sft
SREBP	sterol regulatory element-binding protein	sft
SRF	serum response factor	sft
SRIF	somatotropin release-inhibiting factor (en dash)	sft
SRP	signal recognition particle	sft
SRS-A	slow-reacting substance of anaphylaxis	sft
ss	splice site	sft
SSA-D	succinate semialdehyde dehydrogenase	sft
SSC	standard saline citrate or saline sodium citrate (0.15 M NaCl and 0.015 M sodium citrate = 1× SSC)	OK as is
SSCP	single-strand conformation polymorphism	sft
ssDNA	single-stranded DNA	sft
SSI	solid-state imager	sft
SSPE	saline sodium phosphate EDTA	sft
ssRNA	single-stranded RNA	sft
SST	sea surface temperature (no hyphen)	sft
SSU rRNA	small subunit rRNA	sft
SSV	simian sarcoma virus	sft
STAR proteins	signal transduction and activation of RNA (protein family)	sft
STAT proteins	signal transducers and activators of transcription [specific ones, initial caps (Stat1, Stat2) or all caps (STAT1, STAT2)]	sft
STED	stimulated emission depletion (microscopy)	sft
STEM	scanning transmission electron microscopy, microscope	sft
STM	scanning tunneling microscopy, microscope	sft
STOC	spontaneous transient outward current	sft
STP	standard temperature and pressure	sft
stRNA	small temporal RNA	sft
STRP	short tandem repeat polymorphism	sft
STS	sequence tagged site	sft
STS	scanning tunneling spectroscopy	sft
STZ	streptozotocin	sft
SV40	simian virus 40 (OK as is when spelling it out is a digression from the sentence in which it appears)	sft (if necessary to sense of paper; otherwise OK as is)
SVZ	subventricular zone	sft
SWNT	single-walled carbon nanotube (preferred over SWCNT)	sft
SYK	spleen tyrosine kinase (poster)	sft

T	temperature	sft
T	tritium	sft
$T_{1/2}$	half-life	sft
$t_{1/2}$	half-time	sft
T_3	triiodothyronine	sft
T_4	thyroxine	sft
T_a	ambient temperature	sft
T_c	critical temperature (superconducting transition temperature)	sft
T_F	Fermi temperature	sft
T_g	glass transition temperature	sft
T_H cell	T helper cell (T_H1 , T_H2 , etc.)	sft
T_{FH} cell	T follicular helper cell	sft
T_m	melting transition temperature	sft
T_m	temperature at which 50% of double-stranded DNA or DNA/RNA hybrids are denatured	sft
T_{reg} cell	regulatory T cell (plural: T_{reg} cells)	sft
T state	tense state of enzyme	sft
t test	(statistics)	OK as is
Ta	Taylor number	sft
TAE	tris-acetate-EDTA	sft
TAEA	tris(2-aminoethyl)amine	sft
TAF	TBP-associated factor	sft
TAF _{II250}	(a subunit of TFIID)	OK as is; generally mentioned only where TFIID is mentioned
TAK	transforming growth factor β -activated kinase	sft
TBAP	tetrabutylammonium perchlorate	sft
TBI	total body irradiation	sft
TBOC	<i>tert</i> -butyloxycarbonyl	sft
TBE	tris-borate EDTA	sft
TBE	tris-boric acid-EDTA buffer	sft
TBP	TATA box-binding protein (a subunit of TFIID)	sft
TBS	tris-buffered saline	sft
TBS	theta-burst stimulation	sft
TBW	total body water	sft
TCA	trichloroacetic acid, or tricarboxylic acid	sft (query author)
TCD	thermal conductivity detector	sft
T cell	Texas-type male sterile cell (botany)	sft
T cells	thymus-derived (immunology)	OK as is
TCF/LEF	T cell factor/lymphoid enhancer factor (family of transcription factors)	sft
TCID ₅₀	median tissue culture infectious dose	sft
TCR	T cell receptor (see also T_i)	sft
TCR $\alpha\beta$	$\alpha\beta$ T cell receptor	sft

TDC	tryptophan decarboxylase	sft
T-DNA	transferred DNA	sft
TDP	thymidine 5'-diphosphate	sft
TdR3H	tritiated thymidine	Use [³ H]thymidine instead
TdT	terminal deoxynucleotidyl transferase	sft
TEA-HCl	tetraethylammonium chloride	sft
TEAE-cellulose	<i>O</i> -(triethylaminoethyl)cellulose	sft
TEC	triethylcholine	sft
TEM	transmission electron microscopy, microscope	sft
TES	<i>N</i> -tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid	sft
TFA	trifluoroacetic acid	sft
TFIIA, TFIIB, TFIID, TFIIE, TFIIH	(transcription factors) (TFIID, RNA polymerase II transcription factor D; consists of TBP and about 10 TAFs) (TFIIE, eukaryote-specific transcription initiation factor)	OK as is, if described as transcription factor
TFT	thin-film transistor	sft
TGF- α , TGF α	transforming growth factor- α	sft
TGF- β , TGF β	transforming growth factor- β	sft
TGN	trans-Golgi network	sft
TH	tyrosine hydroxylase	sft
THC	thermohaline circulation	sft
THC, Δ^9 -THC	Δ^9 -tetrahydrocannabinol (radiolabeled form: Δ^9 -[³ H]THC)	sft (follow author on including Δ^9 in acronym)
ThDP	thiamine diphosphate	sft
THF	tetrahydrofuran	sft
Ti plasmid	tumor-inducing plasmid	sft
TIL	tumor-infiltrating lymphocyte	sft
TILM	tunneling-induced luminescence microscopy	sft
TIMP	tissue inhibitor of matrix metalloproteinases	sft
TIMS	thermal ionization mass spectrometry	sft
TIRF	total internal reflection fluorescence (microscopy)	sft
T-J	Triassic-Jurassic (boundary)	sft
TK	thymidine kinase	sft
TL	thermoluminescence	sft
TLC	thin-layer chromatography	sft
TLR	Toll-like receptor	sft
TLS	translesion synthesis	sft
TM	transmembrane	sft
TMA	trimethylamine	sft
TMA	transient molecular anion	sft
TMB	trimethylboron	sft

TMB-4	<i>N,N</i> -trimethylene-bis-(pyridinium-4-aldoxine) dibromide	sft
TMP	thymidine 5'-monophosphate	sft (5'- may be omitted)
TMR	tunneling magnetoresistance	sft
γ -TMT	γ -tocopherol methyltransferase	sft
Tn3, Tn9	transposon 3, transposon 9 (italicize number in abbreviation)	sft
TNF- α , TNF α	tumor necrosis factor- α	sft
TNF- β , TNF β	tumor necrosis factor- β	
TNFR	tumor necrosis factor receptor	sft
TNP	trinitrophenol	sft
TNPP	terrestrial net primary production	
TnT, TnC, TnI	protein components of troponin	sft
TOC	total organic carbon	sft
TOCSY	total correlation spectroscopy	sft
TOF	time-of-flight (mass spectrometer)	sft
tPA	tissue plasminogen activator	sft
TPA	12- <i>O</i> -tetradecanoylphorbol 13-acetate	sft
TPA	tissue plasminogen activator	sft
TPEN	<i>N,N,N',N'</i> -tetrakis(2-pyridylmethyl)ethylenediamine	sft
TP-LEIS	temperature-programmed low-energy ion scattering	sft
TPLSM	two-photon laser scanning microscope, microscopy	sft
TPN	triphosphopyridine nucleotide	sft
TPPE	two-photon photoemission	sft
TRAF	TNF receptor-associated factor	sft
TRAP	telomeric repeat amplification protocol (assay)	sft
TRAP	thyroid hormone receptor-associated protein	sft
TRE	thyroid response element	sft
TRF	thyrotropin-releasing factor	sft
TRF	terminal restriction fragment	sft
TRH	thyrotropin-releasing hormone	sft
trien	triethylenetetramine	sft
tris	tris(hydroxymethyl)aminomethane	OK as is
TRITC	tetramethyl rhodamine isothiocyanate	sft
TRM	thermal remanent magnetization	sft
tRNA	transfer RNA	Define only in abstracts
TRP	transient receptor potential (a channel type)	sft
ts	temperature-sensitive	sft
TSA	trichostatin A (Note: sometimes misspelled tricostatin)	sft
TSH	thyroid-stimulating hormone (thyrotropin)	sft
TSP	total soluble protein	sft
TT	tetanus toxoid	sft
TTC	tetraphenyl tetrazolium chloride	sft

TTF	tetrathiafulvalene	sft
TTTA	1,3,5-trithia-2,4,6-triazapentalenyl	sft
TTX	tetrodotoxin	sft
TUNEL	terminal deoxynucleotidyl transferase–mediated deoxyuridine triphosphate nick end labeling (substitute dUTP if defined earlier)	sft
Tx	thromboxane	sft
TxA ₂	thromboxane A ₂	sft
2DEG	two-dimensional electron gas	sft

UBF	upstream binding factor	sft
UCE	upstream control element	sft
UCP	uncoupling protein	sft
UCS	unconditioned stimulus	sft
UDP	undecaprenol	sft (don't introduce acronym if already in use for the other UDP)
UDP	uridine 5'-diphosphate	sft (5'- may be omitted)
UED	ultrafast electron diffraction	sft
UHV	ultrahigh vacuum	sft
UMP	uridine 5'-monophosphate	sft (5'- may be omitted)
UT	universal time (for example, 9:00 UT)	sft (if used without a number; OK as is if used with a number)
UTC	universal time coordinated	treat as UT
UTL	undifferentiated T cell lymphoma	sft
UTP	uridine 5'-triphosphate	sft (5'- may be omitted)
3'UTR, 5'UTR	3' untranslated region, 5' untranslated region	sft (no space in acronym)
UV	ultraviolet	sft

V	variable region of immunoglobulin chain	sft
V _H	variable region of immunoglobulin heavy chain	sft
V _L	variable region of immunoglobulin light chain	sft
V _e	excluded volume	sft
V _h	holding potential	sft
V _O	void volume	sft
var	(taxonomic term for variant)	Spell out (unless used repeatedly)
VCAM-1	vascular cell adhesion molecule-1	sft
vCJD	variant Creutzfeldt-Jakob disease (the type that affects humans)	sft
VCSEL	vertical cavity surface-emitting laser	sft
V(D)J recombination	(genetics; use parentheses because D elements not always present)	OK as is
VEGF	vascular endothelial growth factor	sft
VEGFR	vascular endothelial growth factor receptor	sft
VIP	vasoactive intestinal peptide	sft
vitamin D		OK as is
VLA-1, -2	very late antigen-1, -2	OK as is
VLBI	very-long-baseline interferometry	sft
VLDL	very-low-density lipoprotein	sft
VLf	very low frequency	sft
VLSI	very-large-scale integration	sft
VMA	vanillylmandelic acid	sft
VNTR	variable number of tandem repeats	sft
VPDB	Vienna Pee Dee belemnite (standard for core sampling)	sft
VPC	vapor-phase chromatography	sft
VRK	vaccinia-related kinase	sft
VSCC	voltage-sensitive Ca ²⁺ channel	sft
VSD	virtually safe dose	sft
VSMOW	Vienna standard mean ocean water	sft
VSV	vesicular stomatitis virus	sft
VSvG	vesicular stomatitis virus glycoprotein	sft
VT-STM	variable-temperature STM	sft
v/v	volume to volume	OK as is
v/w	volume to weight	OK as is

WBCs	white blood cells (usually plural)	sft
WGA	wheat germ agglutinin	sft
WGS	whole-genome shotgun (Celera's genome sequencing method)	sft
WHD	winged-helix domain	sft
WMISH	whole-mount in situ hybridization	sft
WT	wild-type (adjective), wild type (noun)	sft
WTB	wild-type Berlin (flies)	sft
w/v	weight to volume	OK as is
w/w	weight to weight	OK as is

x axis		OK as is
X	Use cap “X” as the single-letter abbreviation for an unspecified or unknown amino acid.	OK as is.
X chromosome		OK as is
x-ray		OK as is; lowercase
		x
XANES	x-ray absorption near-edge structure	sft
XAS	x-ray absorption spectroscopy	sft
XDP	xanthosine 5'-diphosphate	sft (5'- may be omitted)
X-Gal	(a molecular biology reagent) X-galactosidase	sft
XMLD	x-ray magnetic linear dichroism (spectromicroscopy)	sft
XPS	x-ray photoelectron spectroscopy, x-ray photoemission spectroscopy [follow author, or query]	sft
XRD	x-ray diffraction, diffractometer	sft
XRF	x-ray fluorescence	sft
XSCID	X-linked severe combined immunodeficiency	sft
XTP	xanthosine 5'-triphosphate	sft (5'- may be omitted)
XUV	extreme ultraviolet	sft

y axis		OK as is
Y bands	Soret bands (spectroscopy)	sft
Y chromosome		OK as is
Y-labeled	labeled in 3' positions	sft
YAC	yeast artificial chromosome	sft
YAG	yttrium-aluminum-garnet	sft
YAG-Nd laser	yttrium-aluminum-garnet–Nd laser	sft
YBCO	YBa ₂ Cu ₃ O ₇	sft
YEP	1% yeast extract, 1% peptone (medium)	sft
YFP	yellow fluorescent protein	sft
YPD	yeast extract, peptone, and dextrose	sft
yr B.P.	years before the present	sft
	[“calibrated years before the present (cal yr B.P.)” is also permissible – follow authors]	
YSZ	yttria-stabilized zirconia	sft

Z	atomic number	sft
z axis		OK as is
z score	(a statistic)	OK as is
Zap70	(a nonreceptor tyrosine kinase)	OK as is
ZBCP	zero bias conductance peak	sft
ZPM	zero-point motion	sft